NOT MEASUREMENT SENSITIVE

MIL-STD-3050A
1 July 2022

SUPERSEDING
MIL-STD-3050
11 May 2015

DEPARTMENT OF DEFENSE
DESIGN CRITERIA STANDARD

AIRCREW BREATHING SYSTEM (ACBS)

AMSC N/A FSG 15GP

DISTRIBUTION STATEMENT A: Approved for public release. Distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

FOREWORD

1. This standard is approved for use by all Departments and Agencies of the Department of Defense.

2. This standard covers the design, integration, certification, sustainment, and maintenance requirements for an Aircrew Breathing System (ACBS). The intent of this standard is to provide a set of criteria to ensure the safety and optimization of aircrew across all airframe platforms and flying profiles in order to ensure mission success. The expectation is that an ACBS is designed to enable optimal breathing for any given breathing demand, not that the human will acclimate to the ACBS. This standard should be cited in acquisition contracts. An ACBS should be designed to be flexible and adaptable in order to accommodate technologic advances and lessons learned.

3. The changes made in this revision are multifaceted. The title and overall scope of the standard have been changed to include the totality of ACBSs in military aircraft of all Services rather than focusing on one type of ACBS. To this point, sections 3, 4, and 5 have been thoroughly updated to reflect current best practices and guidelines that reflect the requirements for all ACBSs. Unless stated for specific ACBS variants, such as On Board Oxygen Generating Systems (OBOGS) or Liquid Oxygen (LOX), the requirements are general. This standard does NOT cover: parachuting oxygen systems, passenger oxygen systems, cannula delivery systems, medical systems, walk-around oxygen bottles, and full pressure suits. Aircraft Program Offices are the arbiters of specific needs of respective aircraft and any specific issues that are not adequately addressed by this standard should be addressed by the applicable Aircraft Program Office. Additionally, the Aircraft Program Office, as noted in this standard should coordinate with subject matter experts, such as Aerospace Physiologists, Flight Surgeons, Aerospace Engineers, etc., on any issues that arise. This standard should be tailored by the Government as part of the contract. If there are technical or programmatic limitations that make full application of this standard unfeasible, then the program office should coordinate with aeromedical SMEs at the 711th HPW or NAVAIR PMA-202 Human Systems Engineering. If it is not tailored by the Government, the contractor may propose any appropriate tailoring for the applicability to the system and recommend tailoring to the Government for approval.

4. Comments, suggestions, or questions on this document should be addressed to ENGINEERING.STANDARDS@US.AF.MIL. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

CONTENTS

PARAGRAPH **PAGE**

FOREWORD ii
1. SCOPE 1
1.1 Scope. 1
2. APPLICABLE DOCUMENTS 1
2.1 General. 1
2.2 Government documents. 1
2.2.1 Specifications, standards, and handbooks. 1
2.2.2 Other Government documents, drawings, and publications. 2
2.3 Non-Government publications. 2
2.4 Order of precedence. 3
3. DEFINITIONS 3
3.1 Acronym/Abbreviation list. 3
3.2 Definitions. 4
3.2.1 Absorptive Atelectasis. 4
3.2.2 Acceleration Atelectasis. 4
3.2.3 Aviator's Breathing Oxygen. 4
3.2.4 Aircrew Breathing System (ACBS). 4
3.2.5 Auxiliary Breathing Gas System (ABGS). 5
3.2.6 Backup Oxygen System (BOS). 5
3.2.7 Delayed Otic Barotrauma. 5
3.2.8 Emergency Oxygen System (EOS). 5
3.2.9 Explosive decompression. 5
3.2.10 Hypercapnia. 5
3.2.11 Hyperoxia. 5
3.2.12 Hyperpnea. 5
3.2.13 Hyperventilation. 5
3.2.14 Hypocapnia. 5
3.2.15 Hypoventilation. 5
3.2.16 Hypoxia. 6
3.2.17 Insidious decompression. 6
3.2.18 Inspired O_2 Concentration. 6
3.2.19 Minute volume or minute ventilation. 6
3.2.20 Normocapnia. 6
3.2.21 Normoxia. 6

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

3.2.22 Objective. 6
3.2.23 On-Board Oxygen Generator (OBOG). 6
3.2.24 On-Board Oxygen Generating System (OBOGS). 6
3.2.25 Peak Flow. 6
3.2.26 Persistence. 6
3.2.27 Plenum. 6
3.2.28 Pressure Breathing. 7
3.2.29 Primary Breathing Gas System (PBGS). 7
3.2.30 Rapid decompression. 7
3.2.31 Safe-to-fly. 7
3.2.32 Safety critical item. 7
3.2.33 Safety pressure. 7
3.2.34 Threshold. 7
3.2.35 Tidal Volume. 7
3.2.36 Work of Breathing (WOB). 7
4. GENERAL REQUIREMENTS 7
4.1 Background. 7
4.1.1 ACBS oversight, integration, development, acquisition, and verification. 8
4.2 General design concepts. 8
4.3 Design guidelines. 8
4.3.1 ACBS principles. 9
4.4 ACBS architecture. 9
4.4.1 Primary breathing gas system (PBGS). 9
4.4.2 Auxiliary breathing gas system (ABGS). 10
4.5 ACBS materials. 10
4.6 ACBS criticality. 10
5. DETAILED ACBS REQUIREMENTS 11
5.1 ACBS oxygen concentration schedule. 11
5.1.1 Concentration control mode. 11
5.1.2 Max O_2 mode. 11
5.2 ACBS volumetric flow requirements. 13
5.2.1 Transients. 13
5.3 Breathing flow and pressure. 13
5.3.1 Normal operations. 13
5.3.2 Pressure breathing for altitude (PBA). 17
5.3.3 PBG. 18

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

5.3.4 Mask pressure during rapid decompression (see Figure 2b). 20
5.3.5 Regulator outlet safety pressure. 20
5.3.5.1 Safety pressure. 20
5.3.5.2 Emergency safety pressure. 20
5.3.6 Oscillatory activity. 20
5.4 Auxiliary breathing gas system (ABGS). 21
5.4.1 Subsystems. 21
5.4.1.1 Backup oxygen system (BOS). 21
5.4.1.2 Emergency oxygen system (EOS). 22
5.4.1.3 Ejection Oxygen System. 22
5.5 ACBS Integration. 23
5.5.1 Backflow prevention. 23
5.5.2 Breathing gas quality. 23
5.5.2.1 ACBS requirements. 23
5.5.2.2 OBOGS used as PBGS. 23
5.5.3 Temperature. 25
5.5.4 ACBS monitoring, warning, and recording. 25
5.5.5 Advisories, cautions, and warnings. 26
5.5.6 Built-in test functionality (BIT). 28
5.5.7 Materials and components. 28
5.5.8 Human factors. 29
5.5.9 High pressure oxygen storage. 29
5.5.10 Oxygen concentrator waste gas venting. 29
5.5.11 Validation and verification of ACBS. 29
6. NOTES 29
6.1 Intended use. 29
6.2 Acquisition requirements. 29
6.3 Patent notice. 30
6.4 Subject term (key word) listing. 30
6.5 International standardization agreement implementation. 30
6.6. Changes from previous issue. 30
APPENDIX A 31
A.1 SCOPE 31
A.1.1 Background. 31
A.1.2 Perspective. 31
A.2 ACBS MINUTE VENTILATION AND PEAK FLOWS 31

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

A.3 IDENTIFICATION OF PEAK PRESSURES 32
A.3.1 Peak pressures. 32
A.3.1.1 No safety pressure. 33
A.3.1.2 Partial safety pressure. 33
A.3.1.3 Continuous safety pressure. 33
A.3.2 Identification of peak pressures and work of breathing. 33
A.3.2.1 Pressure-volume loop (ACBS with no safety pressure) 34
A.3.2.2 Work of breathing (WOB) and average pressure. 34
A.3.2.3 Magnitude of elasticity. 35
A.3.2.4 Pressure-volume loop (ACBS with partial safety pressure). 35
A.3.2.5 Pressure-volume loop (ACBS with safety pressure). 35
A.4 ADDING UP THE RESPIRATORY LOADS 35
A.4.1 Total respiratory load. 35
A.5 PROCEDURE 36
A.5.1 Parameters to be reported. 36
REFERENCES 36
APPENDIX B 37
B.1 SCOPE 37
B.1.1 Scope. 37
B.2 MEETINGS 37
B.2.1 Design/Redesign meetings. 37
B.2.2 Evaluation criteria. 37
B.3 REVIEWS 38
B.3.1 Engineering technical reviews. 38
APPENDIX C 39
C.1 SCOPE 39
C.1.1 Compliance. 39
C.1.2 Verification testing. 39
C.2 APPLICABLE DOCUMENTS 39
C.2.1 Government documents. 39
C.2.1.1 Specifications, standards, and handbooks. 39
C.2.2 Non-Government publications. 39
C.3 METHODS OF COMPLIANCE 40
C.3.1 System hazard and failure analysis. 40
C.3.2 Qualification. 40
C.3.3 Airworthiness certification. 40

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

C.3.4 Aircraft delivery. 40
APPENDIX D 41
D.1 SCOPE 41
D.2 APPLICABLE DOCUMENTS 41
D.2.1 Other Government documents, drawings, and publications. 41
D.3 METHODS OF COMPLIANCE 41
D.3.1 Digital engineering. 41
D.3.2 Scientific Test and Analysis Techniques (STAT). 41
D.4 QUALIFICATION TESTING 42
D.4.1 Test preparation. 42
D.4.2 Test hardware. 42
D.4.3 ACBS safe-to-fly testing. 42
D.4.3.1 Testing with breathing simulators. 42
D.4.3.2 ACBS chemical contaminant testing (OBOGS equipped aircraft) lab test. 43
D.4.3.3 Testing with humans. 43
D.4.3.4 Safe-to-fly documentation. 44
D.4.3.5 In-aircraft testing. 44
D.4.3.5.1 Ground testing. 44
D.4.3.5.1.1 ACBS chemical contaminant testing (OBOGS equipped aircraft) aircraft ground test. 44
D.4.3.5.2 Flight testing. 44
D.4.4 Flight test report. 44
D.4.5 ACBS CBRN testing. 45
APPENDIX E 46
E.1 SCOPE 46
E.1.1 Scope. 46
E.2 APPLICABLE DOCUMENTS 46
E.2.1 Other Government documents, drawings, and publications. 46
E.3 PROCEDURES 46
E.3.1 Performance check procedures. 46
E.3.1.1 Periodic OBOGS ACBS performance checks. 46
E.3.1.2 Periodic ACBS performance check interval. 46
E.3.1.3 Periodic filter checks. 47
E.3.1.4 OBOGS overhaul. 47
E.3.1.5 Incident where a chemical contaminant is suspected. 47
E.3.1.6 Molecular sieve canister assessment methods. 47
E.3.1.6.1 Molecular sieve canister washout time assessment method. 47

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

E.3.1.6.2 Molecular sieve canister activity assessment method. 48
E.3.1.6.2.1 Molecular sieve canister activity assessment method additional details. 48
E.3.1.6.3 Installed performance testing method. 48
E.3.1.7 Periodic performance checks. 48
E.3.1.7.1 Periodic LOX or GOX ACBS performance checks. 48
E.3.1.7.2 Periodic ACBS performance checks for ACBS other than OBOGS, LOX or GOX. 48
APPENDIX F 50
F.1 SCOPE 50
F.1.1 Scope. 50
F.2 APPLICABLE DOCUMENTS 50
F.2.1 Other Government documents, drawings, and publications. 50
F.3 PROCEDURES FOR CALCULATING THE ALVEOLAR PARTIAL PRESSURE OF OXYGEN 50
F.3.1 Equations. 50
F.3.2 Step-by-step procedures. 51
F.3.2.1 Calculating $PAO2$ at different altitudes. 51
F.3.2.2 Calculating $FIO2$ at different altitudes. 51
F.4 PROCEDURE TO MODIFY THE CAUTION LEVEL IN THE OXYGEN SCHEDULE TO PROTECT DURING RAPID DECOMPRESSION 55
F.4.1 Procedure. 55
F.4.2 Step-by-step procedure. 55
F.4.3 Example. 56
APPENDIX G 58
G.1 SCOPE 58
G.1.1 Scope. 58
G.2 DOCUMENTS USED FOR GUIDANCE 58
G.2.1 Guidance documents. 58
CONCLUDING MATERIAL 60

LIST OF FIGURES

FIGURE 1 ACBS architecture 10
FIGURE 2a Oxygen concentration schedule for aircraft with decompression time greater than one (1) second (such as aircraft with low probability of rapid decompression, e.g., cargo and transport aircraft) 12

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

FIGURE 2b Example of oxygen concentration schedule for aircraft with decompression time less than or equal to one (1) second (such as aircraft with high probability of rapid decompression, e.g., fighter/attack aircraft)……. 12
FIGURE 3 Maximum inhalation and exhalation pressures. Values from the columns labelled for the “No Safety Pressure” (solid lines) and for the “With Safety Pressure (threshold)” (gray interrupted lines) in Table I. Pressures inside the respective lines are acceptable and pressures outside the lines are not…… 14
FIGURE 4 Flow and volume profiles showing how a breath is partitioned into three parts (the numbers illustrate breathing at 5 Gz). Solid lines show one breath. Dotted lines show the beginning of the next breath……... 16
FIGURE 5 Anti-G suit pressure schedule with G……. 18
FIGURE 6 Added mask pressure schedule with PBG. (Schedule also applied to counter-pressure vests if used)…….. 18
FIGURE 7 Example of unacceptable oscillatory activity on the pressure in the mask cavity..…… 19
FIGURE A-1 Peak pressure identification…… 32
FIGURE A-2 Pressure-volume diagram of a breath from an ACBS. No safety pressure (left), partial safety pressure (center), and full safety pressure (right)…….. 33
FIGURE A-3 Pressure-volume diagram with example values…….. 35
FIGURE F-1 Illustration of modified oxygen schedule needed to prevent hypoxia following decompression for aircraft with a cabin differential of 5 psi using a regulator with max pressure of 30 mmHg at 50,000 ft. with pressure onset at 39,000 ft.……. 56

LIST OF TABLES

TABLE I. Maximum inhalation and exhalation peak mask pressure limits and work of breathing (mean pressure) for sinusoidal breathing. 16
TABLE II. Durations of inhalation, pause, and exhalation for simulated AGSM breathing. 17
TABLE III. PBA at start of Inhalation.....18
TABLE IV. ACBS maximum allowable contaminant concentrations in breathing gas. 24
TABLE IV. ACBS maximum allowable contaminant concentrations in breathing gas - Continued. 25
TABLE V. ACBS performance data recording. 26
TABLE VI. Advisory, caution, and warning indications. 27

TABLE A- I. Typical ACBS minute ventilation.....32
TABLE A- II. Instantaneous peak inhalation and exhalation flows for sinusoidal breathing (liters ambient temperature and pressure dry (ATPD) per person). 32

TABLE F- I. Oxygen vs altitude schedule using alveolar gas equation[1/].....53
TABLE F- II. Modified Table F- I oxygen schedule needed to prevent hypoxia following decompression.....57

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

1. SCOPE

1.1 Scope.

This standard establishes requirements for an Aircrew Breathing System (ACBS), incorporating physiological parameters as design criteria for an ACBS used in military aircraft designed for sustained operations below 50,000 ft. In the event that there are requirements not addressed in this standard (such as flight above 50,000 ft.), they should be addressed with the Aircraft Program Office and aerospace physiology experts. The human requirements defined herein should be met by all ACBSs which deliver breathing gas to the masks of aircrew in order to protect them against impaired performance due to adverse physiological states such as hypoxia, hyperoxia, hypercapnia, and hypocapnia. This standard should not presume the aforementioned to be all inclusive. The design criteria should mitigate contamination risk for the crew compartment atmosphere by toxic materials (and should be compatible with equipment or systems used to protect aircrew from chemical warfare agents). ACBSs designs should meet the physiologic standards herein. Integration, certification, and sustainment/maintenance are also addressed. This standard should be cited in acquisition contracts. This standard applies to the acquisition of all systems related to ACBS including, but not limited to new programs, upgrades, and subsystem/component changes. Department of Defense Manual (DoDM) 4120.24, Defense Standardization Program (DSP) Procedures, may be used as a guide for tailoring of standardization documents. This standard should be applied in conjunction with other relevant standardization documents. In the event that there are requirements not addressed in this standard they should be addressed with the Aircraft Program Office. The program office should consult the relevant U.S. Government subject matter experts (SMEs) to include Engineering, Aeromedical, Life Support, Aircrew Flight Equipment, Crew Systems, and Human Systems Integration. Appendix A is a repository of physiological reference information that gives the reader perspective on the basis for ACBS requirements. Appendices B, C, D, and E provide mandatory information. Appendices A, F, and G are advisory.

2. APPLICABLE DOCUMENTS

2.1 General.

The documents listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this standard, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks.

The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

INTERNATIONAL STANDARDIZATION AGREEMENTS

AGARD REPORTS

AGARD-AG-286	Advanced Oxygen Systems for Aircraft John Ernsting and Richard L. Miller (eds.) 1996

(Copies of this document are available online at https://www.sto.nato.int/publications/pages.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

INTERNATIONAL STANDARDIZATION AGREEMENTS

AIR STANDARDS

ASMG 4039 Ed 1 v2	Minimum Physiological Requirements for Aircrew Demand Breathing Systems

(Copies of this document are available online at https://assist.dla.mil.)

ISO STANDARDS

ISO 16900-12	Respiratory protective devices - Methods of test and test equipment - Part 12: Determination of volume-averaged work of breathing and peak respiratory pressures.
ISO/TS 16976-4	Respiratory protective devices - Human factors - Part 4: Work of breathing and breathing resistance: Physiologically based limits

(Copies of these documents are available online at https://www.iso.org/standards.html.)

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-7905	Cylinders, Steel, Compressed Gas, Non-Shatterable, Seamless, 1800 PSI and 2100 PSI
MIL-PRF-27210	Oxygen, Aviator's Breathing, Liquid and Gas
JSSG-2010-10	Crew Systems - Oxygen Systems Handbook

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-882	Department of Defense Standard Practice System Safety

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications.

The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AIR FORCE RESEARCH LABORATORY (AFRL)

AFRL-SA-WP-SR-2016-0018	Handbook of Aerospace and Operational Physiology (HAOP)

(Copies of this document are available online at https://discover.dtic.mil/technical-reports/.)

2.3 Non-Government publications.

The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or in the contract.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

ASTM INTERNATIONAL

ASTM MNL36	Safe Use of Oxygen and Oxygen Systems: Handbook for Design, Operation, and Maintenance

(Copies of this document are available online at http://www.astm.org.)

2.4 Order of precedence.

Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. DEFINITIONS

3.1 Acronym/Abbreviation list.

The following acronyms are applicable to this standard.

ABGS	Auxiliary Breathing Gas System
ABO	Aviator's Breathing Oxygen
ABOS	Automatic Backup Oxygen System
ACBS	Aircrew Breathing System
AFRL	Air Force Research Laboratory
AGSM	Anti-G Straining Maneuver
ATPD	Ambient Temperature and Pressure Dry
BOS	Backup Oxygen System
CBRN	Chemical, Biological, Radiological, Nuclear
CDR	Critical Design Review
CLPM	Cabin Liters per Minute (flow (LPM) at cabin altitude and temperature)
CSMU	Crash Survivable Memory Unit
DOE	Design of Experiments
ECS	Environmental Control System
EOS	Emergency Oxygen System or Ejection Oxygen System
GOX	Gaseous Oxygen
Gz	Acceleration in the z-axis
Hz	Hertz (cycles/second)
inWg	Inches of Water Gauge
L	Liter
LOX	Liquid Oxygen
LPM	Liters Per Minute
mg	Milligram
Min	Minute
mmHg	Millimeters of Mercury
MSL	Mean Sea Level

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

O_2	Oxygen
OBOG	On-Board Oxygen Generator
OBOGS	On-Board Oxygen Generating System
P_AO_2	Alveolar Pressure of Oxygen
P_aO_2	Arterial Pressure of Oxygen
PBA	Pressure Breathing for Altitude
PBG	Pressure Breathing for G
PBGS	Primary Breathing Gas System
PDR	Preliminary Design Review
PF	Protection Factor
P_IO_2	Inspired Partial Pressure of Oxygen
PM	Program Manager
ppmv	Parts Per Million by Volume
psi	Pounds Per Square Inch
psia	Pounds Per Square Inch Absolute
psig	Pounds Per Square Inch Gauge
Sec	Second
SME	Subject Matter Expert
STAT	Scientific Test and Analysis Techniques
WOB	Work of Breathing

3.2 Definitions.

The following definitions are applicable to this standard.

3.2.1 Absorptive Atelectasis.

A condition that is due to the collapse of alveoli in the lung, independent of accelerative forces, resulting in absorption of alveolar gas, due primarily to the loss of nitrogen (nitrogen displacement) within the alveoli resulting from breathing high concentrations of oxygen (greater than 60%).

3.2.2 Acceleration Atelectasis.

A collapse of alveoli in the dependent regions of the lung due to increased accelerative forces. Acceleration Atelectasis is exacerbated by breathing 100% oxygen and, during +Gz exposure, by the use of an anti-G suit. Acceleration atelectasis can cause overt symptoms of chest pain, coughing, and shortness of breath.

3.2.3 Aviator's Breathing Oxygen.

LOX or GOX that meets the requirements of MIL-PRF-27210.

3.2.4 Aircrew Breathing System (ACBS).

The total system within the aircraft that, generates, stores, and delivers breathing gas to the aircrew using the aircrew member's mask. This system includes the Primary and Auxiliary Breathing Gas Supplies (PBGS and ABGS).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

3.2.5 Auxiliary Breathing Gas System (ABGS).

This includes the following or a combination: backup oxygen system, ejection oxygen system, and emergency oxygen system.

3.2.6 Backup Oxygen System (BOS).

An oxygen gas storage system designed to mitigate transient or persistent failure of the primary oxygen delivery ACBS. Sometimes referred to as automatic backup oxygen system (ABOS).

3.2.7 Delayed Otic Barotrauma.

Delayed Otic Barotrauma occurs when the pressure in the middle ear is reduced by absorption of oxygen from gas trapped in the ear. The tissue absorption lowers the gas pressure in the closed space. Pain or injury can result.

3.2.8 Emergency Oxygen System (EOS).

Oxygen storage system designed to immediately operate during an emergency situation such as cabin decompression.

3.2.9 Explosive decompression.

Total loss of cabin pressurization in less than one-tenth (0.1) second.

3.2.10 Hypercapnia.

Excessive arterial partial pressure of carbon dioxide (also referred to as hypercarbia). This can be due to rebreathing of exhaled gas or hypoventilation (PaCO2 above 45 mmHg or 6.0 kPa).

3.2.11 Hyperoxia.

An excessive level of oxygen that may cause physiological degradation, resulting in impairment of either both function and exposure of cells or individually, tissues and organs to a higher partial pressure than normal.

3.2.12 Hyperpnea.

Increased minute ventilation that matches increased metabolic needs of the body. Normocapnia is maintained. Example: Hyperpnea due to exercise. Increased ventilation can be caused by increased rate and/or depth of breathing.

3.2.13 Hyperventilation.

A minute ventilation that exceeds the metabolic needs of the body. Hyperventilation results in hypocapnia (decrease in the arterial partial pressure of carbon dioxide).

3.2.14 Hypocapnia.

Abnormally low arterial partial pressure of CO_2 (<35 mmHg, <4.7 kPa), caused by hyperventilation.

3.2.15 Hypoventilation.

An inadequate minute ventilation resulting in an increased amount of carbon dioxide in the blood.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

3.2.16 Hypoxia.

Arterial hypoxia is defined as PaO2 less than 80 mmHg, or 10.7 kPa. During hypoxia the supply of oxygen to tissue, particularly to the brain, is insufficient, resulting in impairment of function.

3.2.17 Insidious decompression.

Loss of cabin pressurization over a period of ten (10) seconds or more, which often is undetected by aircrew.

3.2.18 Inspired O_2 Concentration.

Breathing gas O_2 concentration by volume delivered to the aircrew member's mask.

3.2.19 Minute volume or minute ventilation.

The volume of breathing gas used per aircrew member averaged over a one (1) minute time span. Equal to the aircrew member's tidal volume multiplied by the breaths per minute.

3.2.20 Normocapnia.

Normal arterial partial pressure of carbon dioxide (40 ±2 mmHg, 5.3 ±0.3 kPa).

3.2.21 Normoxia.

Alveolar partial pressure of oxygen (PAO_2) in the range of 100 to 104 mmHg, 13.3 to 13.9 kPa. This represents breathing air at sea level.

3.2.22 Objective.

Desired performance that exceeds the Threshold.

3.2.23 On-Board Oxygen Generator (OBOG).

A component used to produce breathing gas; generally referred to as an oxygen concentrator.

3.2.24 On-Board Oxygen Generating System (OBOGS).

A type of PBGS that includes an oxygen concentrator.

3.2.25 Peak Flow.

Maximum flow during inhalation or exhalation or similarly generated by a breathing machine. If flow is sinusoidal, peak flow is the amplitude of the function.

3.2.26 Persistence.

The time required for a given caution, warning, or advisory condition to exist before it is annunciated to the aircrew.

3.2.27 Plenum.

A reservoir that contains gas only. Its function is to allow for gas mixing and to dampen pneumatic pressure swings.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

3.2.28 Pressure Breathing.

The technique of breathing when oxygen is supplied directly to an individual at a pressure higher than the ambient pressure.

3.2.29 Primary Breathing Gas System (PBGS).

This includes the concentrator or storage container, where applicable, the breathing regulator, supply hoses/tubes, mask, and any and all equipment between the primary oxygen supply and the mask.

3.2.30 Rapid decompression.

Total loss of cabin pressurization in one (1) second (+0.0 / -0.2 seconds).

3.2.31 Safe-to-fly.

Certification that an aircraft ACBS safely meets the aeromedical and physiological requirements of the aircrew.

3.2.32 Safety critical item.

A hardware or software item, the failure of which has been determined through analysis to potentially contribute to a hazard with catastrophic or critical mishap potential (e.g., safety-critical function, safety-critical path, and safety-critical component), or an item that may be implemented to mitigate a hazard with catastrophic or critical mishap potential (see MIL-STD-882).

3.2.33 Safety pressure.

A breathing regulator output pressure slightly greater than ambient, typically 1.0 to 1.5 inWg. The purpose is to provide elevated pressure in the mask to prevent cabin air from leaking into the mask.

3.2.34 Threshold.

The minimum acceptable values for a requirement.

3.2.35 Tidal Volume.

The volume of breathing gas in one breath.

3.2.36 Work of Breathing (WOB).

The amount of physical work used to overcome the breathing resistance in the ACBS, expressed as volume-averaged pressure (see Appendix A for details). WOB in this standard refers to the additional work imposed by the ACBS.

4. GENERAL REQUIREMENTS

4.1 Background.

The criteria established in this military standard are applicable to all ACBSs. The crew shall be continuously supplied with physiologically appropriate breathing gas during all phases of aircraft operation whether on the ground or in-flight, including during any ECS and engine bleed air anomalies, transients or shutdown. The intent of these criteria is to address normal and emergency operations. The ACBS, which supplies breathing gas to aircrew shall protect against

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

hypoxia, at all cabin altitudes and shall also protect against the inhalation of toxic materials such as fuels, lubricants, and products of thermal combustion which may contaminate the cabin environment and ECS. The ACBS shall also provide safe breathing gas in the case of smoke and fumes in the cockpit, and the breathing gas shall be free from smoke and fume contamination. The ACBS shall also integrate into the aircraft's CBRN protection system. The pressure, flow, composition, and volume of the inspired gas delivered to the aircrew member and the work of breathing must be within the requirements established in this document. The ACBS should be optimized to the extent possible to maximize cognitive and physical performance required to aviate, navigate, and communicate in the tactical environment.

4.1.1 ACBS oversight, integration, development, acquisition, and verification.

The execution of oversight and integration of the ACBSs are currently managed by the Services' Aircraft Program Offices. Development, acquisition, and verification of ACBS and its subsystems and related systems (such as masks, helmets, etc.) shall be coordinated by the Program Offices with the relevant DoD subject matter experts (SMEs) to include Engineering, Aeromedical, Life Support, Aircrew Flight Equipment, Crew Systems, and Human Systems Integration. (e.g. 711th HPW, AFLCMC/EZFC, AFLCMC/WNU, PMA 202, NAWCAD HSED, NAMRU-D, USAARL).

4.2 General design concepts.

The ACBS should be designed to achieve the following to the greatest degree possible:

a. Meet aircrew respiratory demands to maintain physiologically sufficient levels of pulmonary oxygen and carbon dioxide, facilitate normal pulmonary function, and minimize work of breathing during in all phases of flight.

b. Minimize degradation of cognition, physical performance, or aircrew communication.

c. Prevent hypoxia without inducing or exacerbating acceleration atelectasis, absorption atelectasis or delayed otic barotrauma; and following decompression of a pressurized cabin up to the maximum altitude the aircraft is designed to fly.

d. Prevent hypercapnia and hypocapnia while the cabin is pressurized, and following decompression of the pressurized cabin at maximum altitude the aircraft is designed to fly.

e. Prevent induction of environmental gases into the mask including smoke/fumes, and CBRN hazards.

f. Provide pressure breathing automatically when required.

g. Provide aircrew the ability to manually select/deselect safety pressure and emergency safety pressure.

h. Prevent or minimize significant oscillations of pressure, flow, or oxygen concentration at the aircrew member's mask.

i. Protect against adverse pulmonary and hypoxia related consequences caused by rapid cabin depressurization or ejection.

j. Provide pilot controllable ACBS manual interfaces, i.e., selectable safety pressure, composition control, and "MAX" or "100%" operating mode.

4.3 Design guidelines.

The aviator's performance flying the aircraft under dynamic conditions should be optimized. Optimization requires that the ACBS supports aircrew physiologic needs throughout the flight

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

profile in all conditions that the aircraft is designed to (such as operation in CBRN environments if required). Essential to this concept, hypoxia should be avoided; stable pressure, flow, volume, and oxygen partial pressure should be delivered; work of breathing (WOB) should be minimized; and the ACBS should not impair aircrew performance. The ACBS should optimize aircrew performance. To this end, when ACBS requirements are in conflict with other requirements, the contractor shall coordinate with the Aircraft Program Office. The Program Office should consult with either the Air Force or Navy Aerospace Physiology SMEs.

4.3.1 ACBS principles.

The requirements in this standard were established using the principles below for guidance.

a. Avoid hypoxia and maintain blood oxygen levels by maintaining inspired oxygen concentrations as provided in the tables and figures of this Standard, for altitude, pressure, and workload, resulting in a blood oxygen saturation above 95%. Maintain at least 30 mmHg alveolar oxygen partial pressure after a rapid decompression.

b. Achieve a stable inspired oxygen concentration.

c. Avoid excessive WOB.

d. Avoid physiologically significant hyperoxia.

e. Avoid physiologically significant hypocapnia and hypercapnia.

f. Avoid abnormal states such as atelectasis.

g. Ensure the quality of the breathing gas.

h. Provide fault detection.

i. Provide redundancy in the oxygen source as a failure-mode mitigation.

j. Optimize ACBS reliability, availability, and maintainability.

k. Use an open-architecture to improve ACBS performance and optimize aircrew performance, i.e., facilitate technology innovation including but not limited to, integration of emerging sensor technology for ACBS feedback loops.

4.4 ACBS architecture.

The ACBS shall consist of a primary source of breathing gas, which is augmented by auxiliary sources as necessary for the aircraft application (see Figure 1).

4.4.1 Primary breathing gas system (PBGS).

The ACBS shall provide physiologically appropriate breathing gas during all operational environments as specified by the Aircraft Program Office.

a. The Primary Breathing Gas System (PBGS) source may be high-pressure gaseous oxygen (GOX), liquid oxygen (LOX), or be produced using an on-board oxygen generator/concentrator.

b. If the PBGS does not consist of LOX or GOX, then an Auxiliary Breathing Gas System shall also be required.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

4.4.2 Auxiliary breathing gas system (ABGS).

The ABGS supplements all non-LOX/GOX systems to continuously meet breathing gas requirements during transient disruptions of the PBGS, certain in-aircraft emergencies, and ejection requirements.

a. The ABGS shall meet the requirements for the following (see 5.4 for details):

(1) Backup Oxygen System (BOS) – Breathing gas available to supplement the PBGS during transient periods of disrupted performance.

(2) Emergency Oxygen System (EOS) – Breathing gas immediately available for rapid decompression and failure of the PBGS.

(3) Ejection Oxygen System – Breathing gas automatically activated during ejection of the aircraft (for ejection seat equipped aircraft).

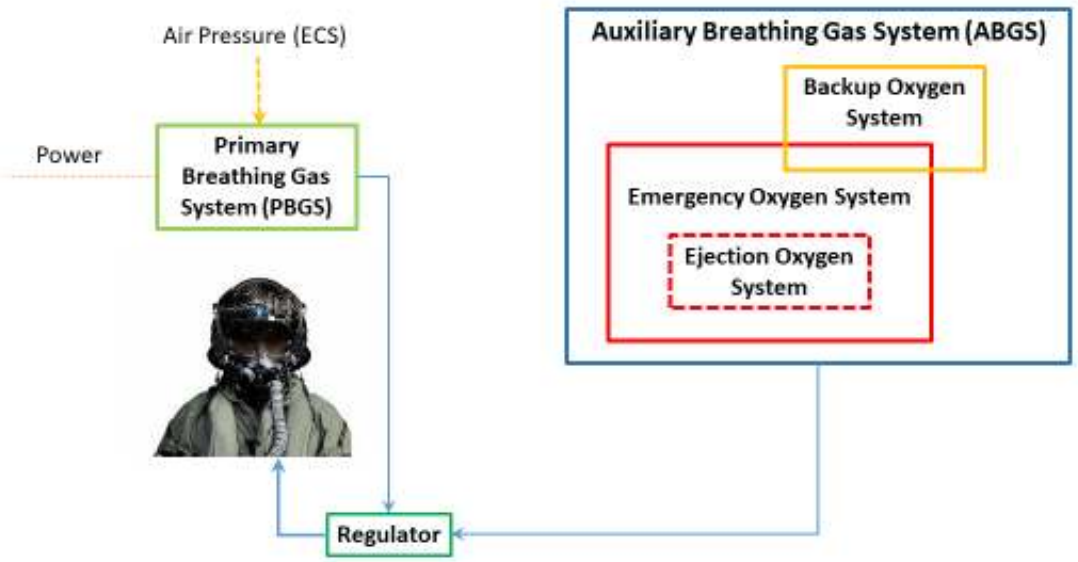


FIGURE 1. ACBS architecture.

4.5 ACBS materials.

Use of light metals (i.e., aluminum, magnesium, titanium, etc.) as an oxygen-wetted material in an oxygen system should be made with caution (see ASTM MNL36). While titanium and magnesium should almost never be used, aluminum alloys have been used successfully in some oxygen applications. However, the use of flammable materials requires mitigation due to the extreme flammability and ignitability of light metals materials in pressurized oxygen systems. The combination of light metals and pressurized oxygen can combine to create catastrophic fires if active ignition mechanisms are not mitigated. If practical, the use of some copper and nickel based metals such as brass, or Monel will significantly reduce the risk and consequence of ignition and burning.

4.6 ACBS criticality.

Aircraft Program Offices shall manage ACBSs as a safety-critical system in accordance with MIL-STD-882 and shall be managed throughout the aircraft's lifecycle to ensure optimum aircrew performance and mission effectiveness.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

5. DETAILED ACBS REQUIREMENTS

5.1 ACBS oxygen concentration schedule.

The ACBS shall have at least two (2) modes of operation: a Concentration Control mode and a Max O_2 mode.

5.1.1 Concentration control mode.

Sometimes called “Normal” or “Concentration control” mode, this mode requires the oxygen concentration from the ACBS be controlled and scheduled according to cabin altitude. In Concentration Control mode, the ACBS shall provide breathing gas and the caution and warning functions to conform to Figure 2a for aircraft with low probability of rapid decompression, e.g., cargo and transport aircraft. However, the oxygen concentration schedule shown on Figure 2a does not take into account the need to maintain adequate aircrew partial pressure of oxygen in the event of a rapid loss of cabin pressurization. Figure 2b shows the oxygen concentration schedule to express “the minimum concentration required in relation to cabin altitude to prevent hypoxia in steady state and in the event of a rapid decompression in an aircraft with a 5 psi differential pressure cabin and using a pressure of 30 mmHg at 50,000 feet” (see Ernsting 1996). See Appendix F for instructions for calculating the oxygen concentration schedule and caution and warning requirements. See Tables F-I and F-II for additional oxygen concentration vs altitude schedule data. Aircraft Program Offices can contact the USAF 711th Human Performance Wing (711 HPW) for assistance at 711.hpw.rh.workflow@us.af.mil or NAVAIR PMA 202 NAWCAD HSD NAWCAD_HSED_MAIL@navy.mil.

a. The contractor shall define the breathing gas oxygen concentration target line (measured at the inlet to the O_2 mask) that meets the following parameters:

   (1) Target line must be within the design zone bounds as demonstrated by Figures 2a and 2b.

   (2) The allowable variation from the concentration target line shall be within ±5% O_2 at least 85% of the operational time, and within ±10% O_2 at least 98% of the operational time. Furthermore, excursions outside the ±5% target line shall be no longer than one (1) consecutive minute. Oxygen delivery shall be above the lower boundary of the design zone as labeled (see Figures 2a and 2b) 100% of the operational time.

b. The ACBS shall provide Pressure Breathing for Altitude (PBA) for cabin altitudes as depicted on Figures 2a and 2b and specified in Table III.

c. The oxygen concentration schedule and caution and warning functions shall be tailored for aircraft not using the standard 5 psi pressurization schedule or having a maximum ceiling less than 50,000 feet.

5.1.2 Max O_2 mode.

Sometimes referred to as “100%”, although actually the maximum O_2 that the ACBS can deliver, this mode requires the ACBS to provide higher concentrations of oxygen when selected by the aircrew for emergency situations. Max O_2 mode shall always be available for pilot selection while PBGS is providing breathing gas.

a. When max mode is active, the ACBS shall provide breathing gas oxygen concentration as follows:

   (1) Below 8,000 feet MSL, Max O_2 mode shall provide >70% O_2 concentration.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

(2) From 8,000 feet to 25,000 feet MSL cabin altitude, Max O_2 mode shall provide ≥80% O_2 concentration.

(3) Above 25,000 feet MSL cabin altitude, Max O_2 mode shall provide ≥94% O_2 concentration.

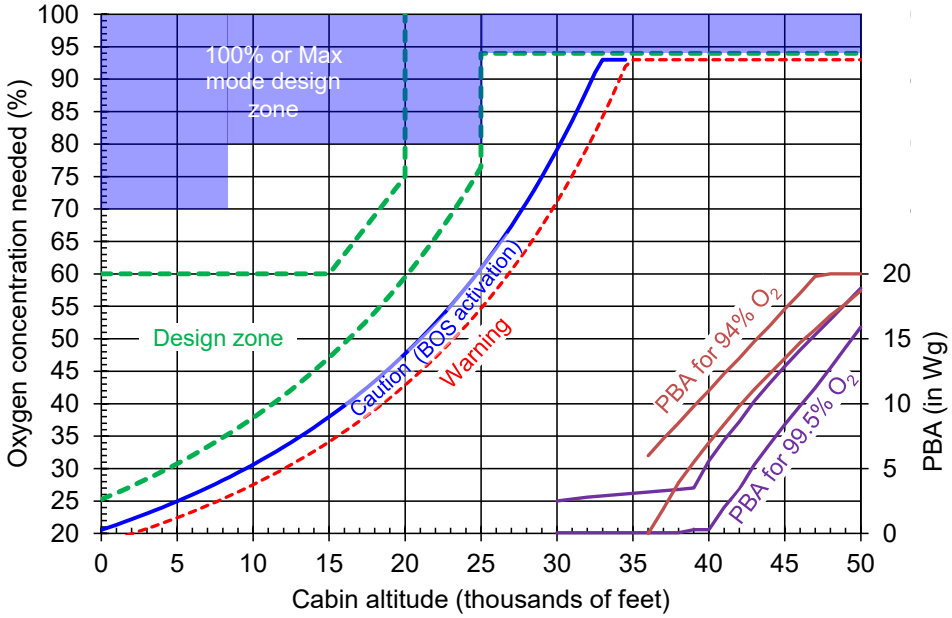


FIGURE 2a. Oxygen concentration schedule for aircraft with decompression time greater than one (1) second (such as aircraft with low probability of rapid decompression, e.g., cargo and transport aircraft).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

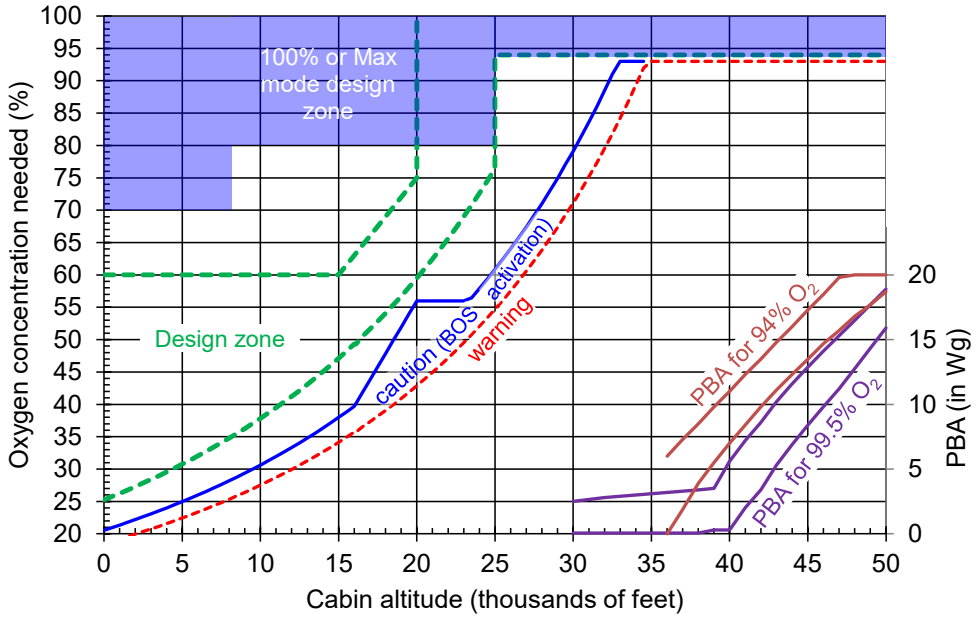


FIGURE 2b. Example of oxygen concentration schedule for 5 psi aircraft with decompression time less than or equal to one (1) second (such as aircraft with high probability of rapid decompression, e.g., fighter/attack aircraft).

5.2 ACBS volumetric flow requirements.

The ACBS shall provide sufficient volumetric flow to meet aircrew-breathing demands for the aircraft application. The system shall be designed to meet the peak flow levels (for each person) in Table I. Only passengers are considered non-crewmembers. Typical flow volumes for various flight conditions are shown in Appendix A, Table A-I. For additional information, aeromedical pulmonary medicine experts should be consulted. For aircraft that use an OBOGS as the PBGS, the requirements must be met at the minimum appropriate OBOGS supply pressure.

5.2.1 Transients.

Performance of the ACBS shall not be degraded by transients from the powerplant, ECS, and electrical system, and the ACBS shall still meet the requirements specified in 5.3, 5.4, and 5.5. Examples of transients are: engine bleed air and ECS pressure transients, pressure fluctuations, rapid pressure drop-offs, and pressure spikes. An example of how to mitigate OBOGS supply pressure transients would be to provide redundant inlet air supply lines to the OBOG.

5.3 Breathing flow and pressure.

The following details the requirements for the breathing flow and pressure.

5.3.1 Normal operations.

The ACBS shall be capable of achieving peak and average inhalation and exhalation flows as described in Table I. The ACBS shall ensure an adequate volume of breathing gas is available at high breathing demands and meet the mask pressure requirements noted on Figure 3 (without safety pressure) and in Table I. If OBOGS is used as the PBGS, a plenum in the product gas line is often used to improve breathing dynamics. Prior applications have used plenums with volumes of 100-250 in^3 per aircrew member. The plenum volume, however, should be sized and optimized based on OBOGS performance, aircrew needs during all flight conditions, and aircraft

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

performance including bleed air pressures, climb rates, and number of crew. If a product gas plenum is used it shall have a void volume. A molecular sieve filled plenum shall not be used.

a. Mask pressures and Work of Breathing shall not exceed the threshold values in Table I during relevant operational conditions. The maximum inhalation and exhalation peak pressures are shown on Figure 3. From an aeromedical perspective, optimal aircrew protection against dilution of oxygen at altitude, exposure to contaminants (smoke and fumes), and CBRN would be provided by designing to the objective column.

   (1) Ground operations: test at test conditions 1 and 2.

   (2) Flight operations: test at test conditions up to number 4.

   (3) Dynamic maneuvering: test at test conditions up to number 10.

   (4) AGSM: test at test conditions 11-13 based on aircraft G rating.

b. The values in Tables I represent the most conservative value of mask cavity pressures at 30, 90, 150, and 200 LPM ATPD (see Air Standard ASMG 4039 Ed 1 v2 for guidance), with the addition of mask cavity pressure at 258 LPM ATPD, and the values from Shykoff and Warkander (2011). Appendix A gives guidance on how respiratory loads (e.g., flow resistance, safety pressure) add up.

c. The breathing profiles shown in Table II are described on Figure 4. One complete breath is drawn using a solid line while dotted lines show the last part of the previous breath and the beginning of the next. The duration of the entire breath is labelled as T_{tot}. The duration of the inspiration is labelled as T_{in}. The duration of the end-inspiratory pause that follows is labelled T_{pause}. The duration of the expiration is shown as T_{ex}.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

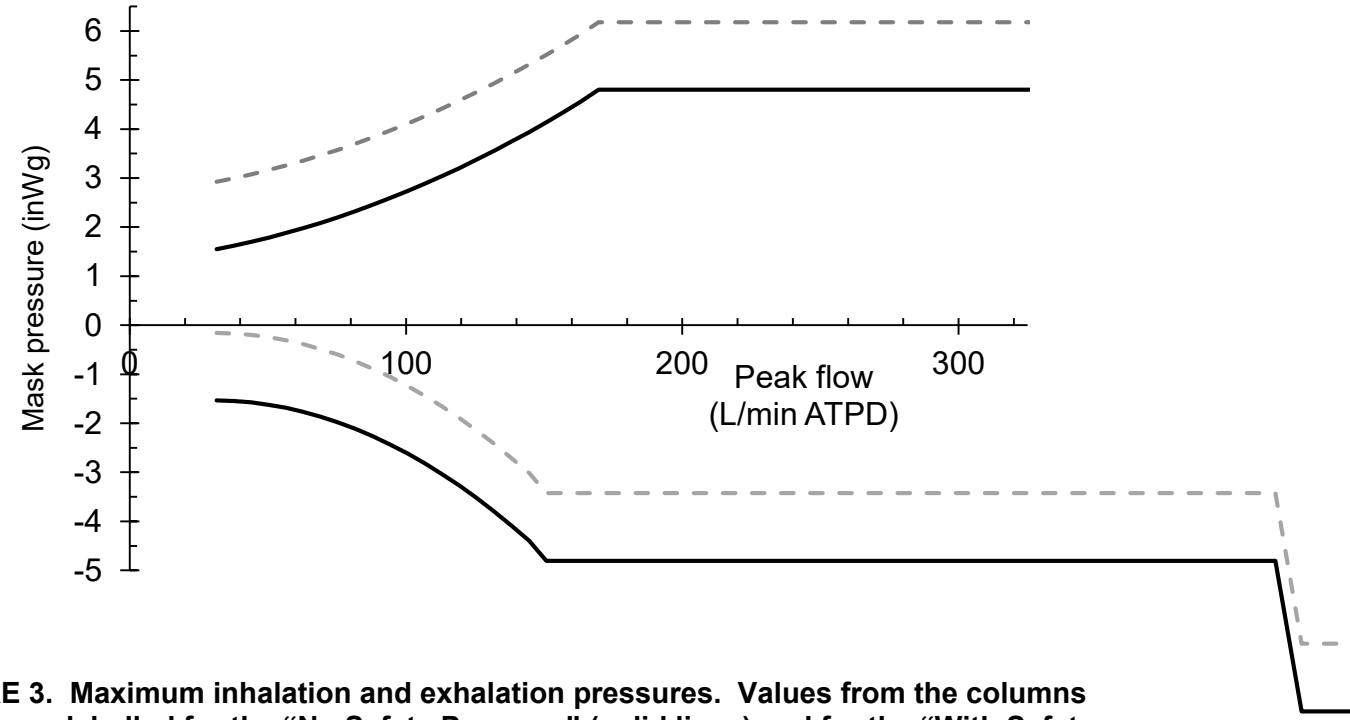


FIGURE 3. Maximum inhalation and exhalation pressures. Values from the columns labelled for the "No Safety Pressure" (solid lines) and for the "With Safety Pressure (threshold)" (gray interrupted lines) in Table I. Pressures inside the respective lines are acceptable and pressures outside the lines are not.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

TABLE I. Maximum inhalation and exhalation peak mask pressure limits and work of breathing (mean pressure) for sinusoidal breathing.

					Mask Pressure Requirements (Referenced to cabin pressure)							
					No Safety Pressure (threshold)		With Safety Pressure (threshold)		With Safety Pressure (objective)		WOB/V_T [1/] mean pressure (threshold)	
Test Condition	Minute Ventilation (L/min)	Peak Flow [2/] (L/min)	Tidal Volume (L)	Breathing Frequency (breaths per minute)	In (in Wg)	Ex (in Wg)	In (in Wg)	Ex (in Wg)	In (in Wg)	Ex (in Wg)	In (in Wg)	Ex (in Wg)
1	10.0	31	1.0	10.0	-1.5	1.5	-0.1	2.9	0.1	2.9	1.2	1.2
2	20.0	63	1.0	20.0	-1.8	2.0	-0.4	3.4	0.0	3.4	1.5	1.5
3	30.0	94	1.5	20.0	-2.3	2.5	-0.9	3.9	0.0	3.9	1.9	1.9
4	35.0	110	1.5	23.3	-2.9	3.0	-1.5	4.3	0.0	4.3	2.2	2.2
5	40.0	126	1.0	40.0	-3.5	3.4	-2.2	4.8	0.0	4.8	2.5	2.5
6	50.0	157	2.0	25.0	-4.8	4.3	-3.4	5.7	0.0	5.7	3.3	3.3
7	60.0	188	2.5	24.0	-4.8	4.8	-3.4	6.2	0.0	6.2	3.6	3.6
8	65.0	204	2.0	32.5	-4.8	4.8	-3.4	6.2	0.0	6.2	3.6	3.6
9	75.0	236	1.5	50.0	-4.8	4.8	-3.4	6.2	0.0	6.2	3.6	3.6
10	75.0	236	2.5	30.0	-4.8	4.8	-3.4	6.2	0.0	6.2	3.6	3.6
11[3/]	41.6	180	1.6	26.0	-4.8	4.8	-3.4	6.2	0.0	6.2	3.6	3.6
12[3/]	52.5	258	1.75	30.0	-4.8	4.8	-3.4	6.2	0.0	6.2	3.6	3.6
13[3/]	34.5	322	1.5	23.0	-4.8	4.8	-3.4	6.2	0.0	6.2	3.6	3.6

Maximum elastance: 1 kPa per L (4.0 inWg per L)[1/].

The negative starting pressure (point A on Figure A-2) shall not be below -1 kPa (-4.0 inWg)[1/].

The positive starting pressure (point A on Figure A-2) shall not exceed +1 kPa (+4.0 inWg)[1/].

1/	Derived from ISO 16976-4.
2/	Calculated as π x minute ventilation (sinusoidal breathing pattern) unless otherwise noted.
3/	For non-PBG applications. Breathing profiles as described in Table II and on Figure 4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

TABLE II. Durations of inhalation, pause, and exhalation for simulated AGSM breathing.

Test Condition	Aircraft G rating	Inhalation Duration (%)	Pause (%)	Exhalation Duration (%)
11	5	36	28	36
12	7	32	36	32
13	9	18	64	18
See Figure 4 for waveform example.				

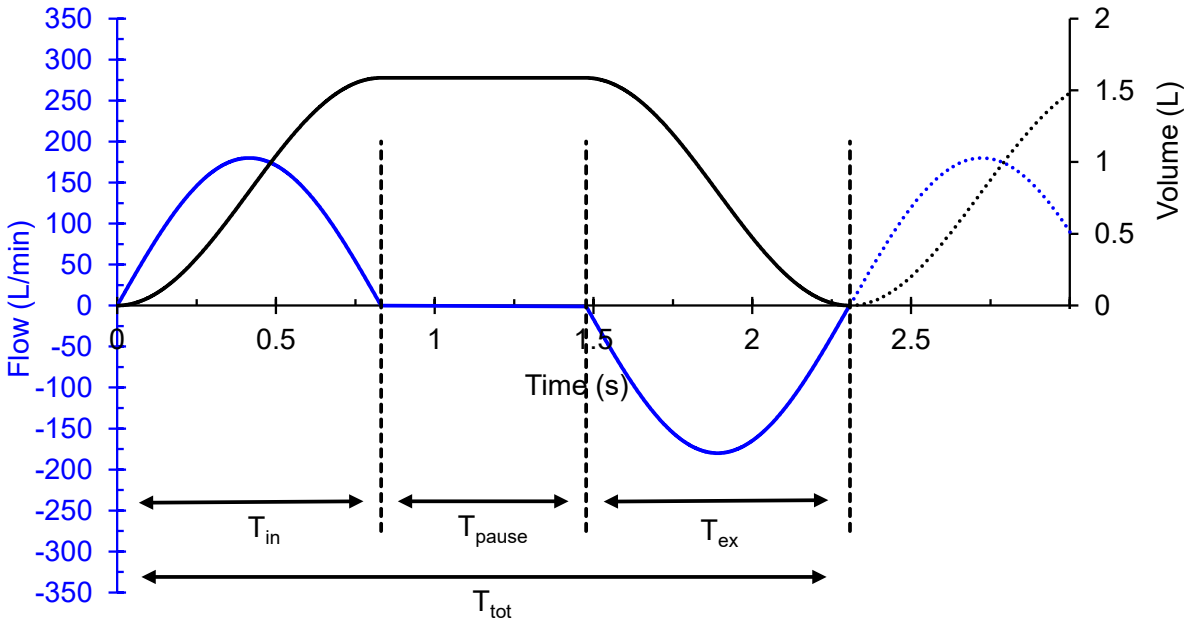


FIGURE 4. Flow and volume profiles showing how a breath is partitioned into three parts (the numbers illustrate AGSM breathing at 5 Gz). Solid lines show one breath. Dotted lines show the beginning of the next breath.

5.3.2 Pressure breathing for altitude (PBA).

Sustained operations at high altitude (>43,000 feet cabin altitude) require a full or partial pressure suit. For exposure to high altitude decompression without a pressure suit, the ACBS shall provide PBA shown in the schedule in Table III, up to the aircraft ceiling altitude.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

TABLE III. PBA at start of Inhalation.

Cabin Altitude (thousands of feet)	Regulator outlet pressure (inWg) when using 99.5% oxygen[1/]	Regulator outlet pressure (inWg) when using 94% oxygen[1/]
30	0.01 to 2.5	
32	0.01 to 2.8	
34	0.01 to 3.0	
36	0.01 to 3.2	0.0 to 6.0
38	0.01 to 3.4	3.9 to 8.5
40	0.3 to 5.6	7.0 to 11.0
42	3.4 to 8.6	9.8 to 13.5
44	6.9 to 11.6	12.3 to 16.0
46	9.8 to 14.1	14.7 to 18.5
48	12.8 to 16.5	16.8 to 20.0
50	15.9 to 18.9	18.7 to 20

1/ Using Test Condition 6 in Table I, for at least two minutes.

NOTE: Safety pressure, if used, may meet the PBA requirement at low altitudes.

5.3.3 PBG.

If PBG is utilized, the following requirements shall be applied:

a. PBG shall be integrated with the aircraft anti-G system.

b. Any Anti-G valve schedule other than what is shown on Figure 5 shall be approved by the program office.

c. PBG mask pressures (pressure in absence of flow) shall be in accordance with Figure 6. PBG shall automatically activate at +4 Gz. The breathing pressure shall then increase by 12 ±4 mmHg (6.42 ±2.1 inWg) with each +1 Gz from 4 to +9 Gz

d. There shall be a fail-safe design to prevent PBG without G-suit inflation (a failure mode that could cause a physiological incident).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

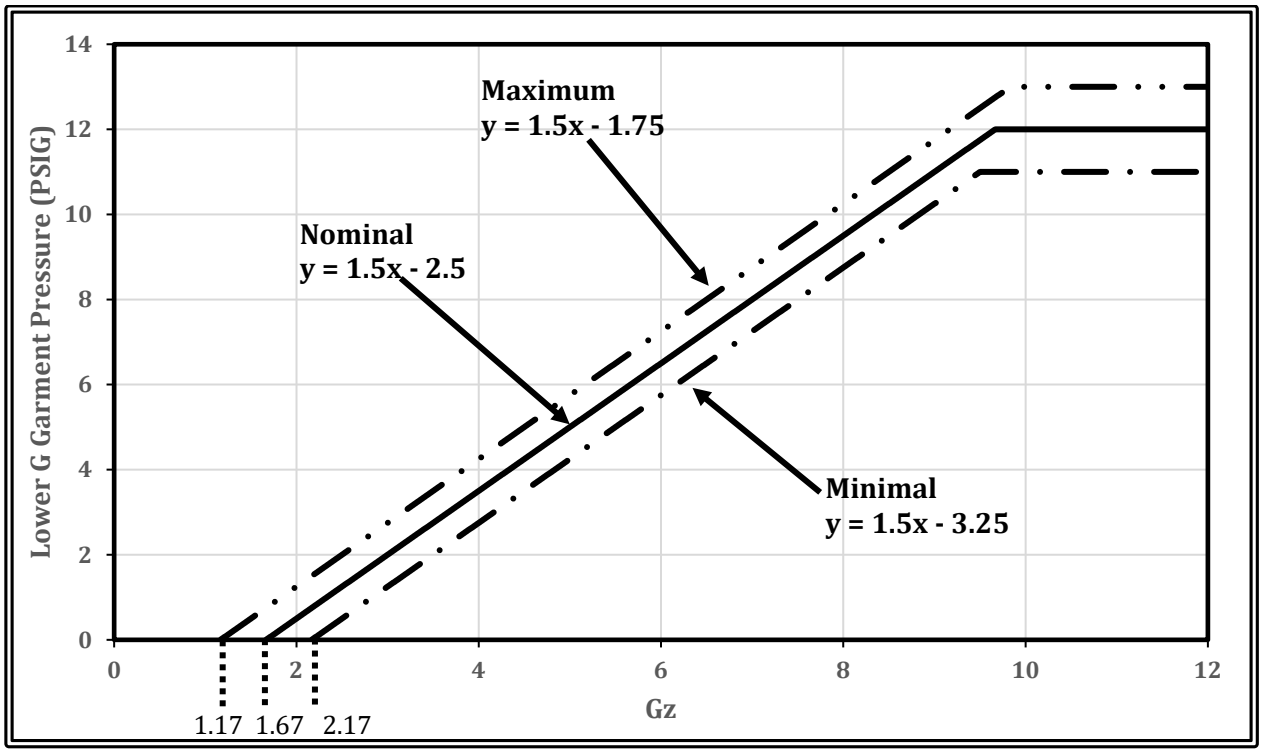


FIGURE 5. Anti-G suit pressure schedule with G.

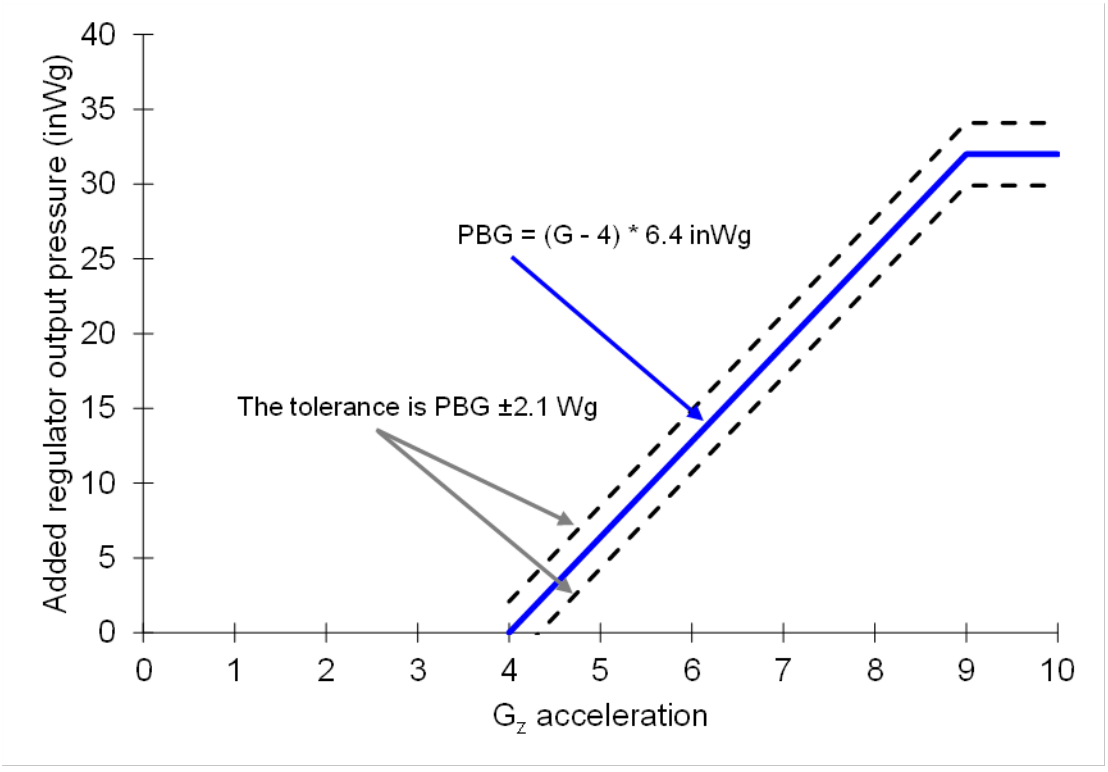


FIGURE 6. Added mask pressure schedule with PBG. (Schedule also applies to counter-pressure vests if used.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

5.3.4 Mask pressure during rapid decompression (see Figure 2b).

The pressure in the mask cavity during and immediately after a rapid decompression (5 psid in one [1] second or less) shall not exceed the cabin pressure plus 22 inWg (41 mmHg) for longer than 100 milliseconds.

The mask pressure shall never exceed 37.5 inWg (70 mmHg) above the cabin pressure. If the expected flight profile is expected to exceed 50,000 feet, then a Counter-pressure vest is required.

5.3.5 Regulator outlet safety pressure.

5.3.5.1 Safety pressure.

If included in the design, shall be aircrew selectable as either “off” or “on”.

a. “Off” mode shall result in a mask pressure of 0 to 0.2 inWg when no breathing is applied to the mask. When safety pressure is off, the mask shall not free-flow when removed from the aircrew member’s face.

b. In absence of flow, “On” mode shall provide a safety pressure of 0.5 to 1.5 inWg. When safety pressure is selected, either manually or automatically, the pressure in the mask cavity shall meet the requirements in Table I.

5.3.5.2 Emergency safety pressure.

a. “Emergency” mode, when selected, shall provide a safety pressure of 3 to 4 inWg for emergency protection (such as for cabin air contaminants, smoke or fumes). The pressure in the mask cavity shall be greater than that of the cabin pressure at inhalation flows of up to at least 120 LPM (ATPD) peak sinusoidal flow.

5.3.6 Oscillatory activity.

The ACBS shall supply mask pressures such that the peak-to-peak heights of any oscillations (repeated pressure fluctuations) shall not exceed 0.25 inWg, at a frequency ≥4 Hz. Figure 7 shows an example, with the height of the oscillation indicated by the double headed arrow. If initial testing using a breathing machine reveals oscillatory activity outside this limit, then further testing with human subjects shall be conducted to rule out machine generated oscillations (see Appendix A for more information).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

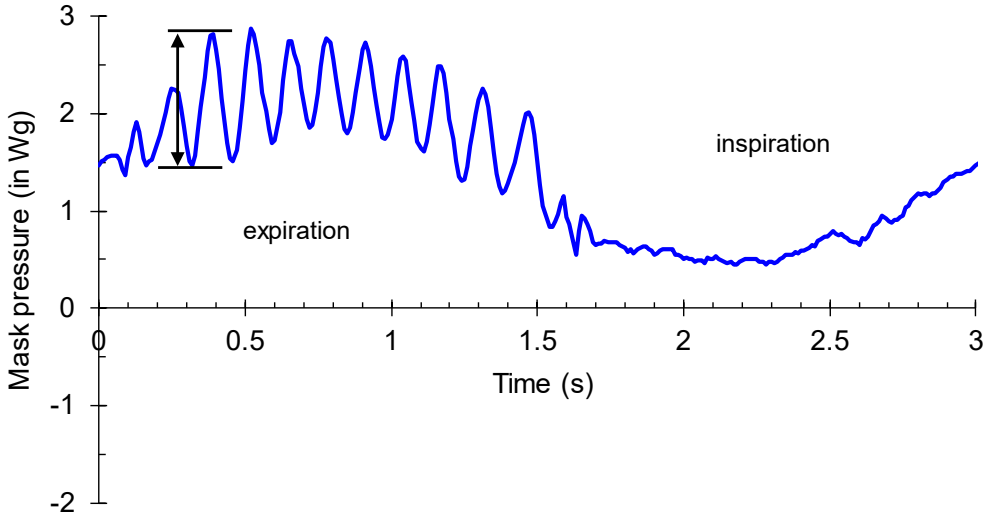


FIGURE 7. Example of unacceptable oscillatory activity on the pressure in the mask cavity.

5.4 Auxiliary breathing gas system (ABGS).

The ABGS shall provide for the following functions: backup oxygen, emergency oxygen, and ejection oxygen.

5.4.1 Subsystems.

Separate subsystems (BOS, EOS, Ejection Oxygen System) may be used or combined, provided the following sets of requirements are met.

a. All ABGS breathing gas shall go through a breathing regulator. Constant flow or orifice controlled flow systems shall not be used.

b. All breathing gas provided by ABGS shall not exceed 7 ppm of water vapor or a maximum dew point of -82 °F.

All breathing gas provided by an ABGS shall meet the contamination requirements in Table IV.

5.4.1.1 Backup oxygen system (BOS).

The BOS is used to provide supplemental breathing gas during transient loss in performance from the PBGS. The system used for BOS shall provide 90% or greater oxygen.

a. The BOS shall be *auto-activated* upon any PBGS failure that can interrupt or degrade the breathing gas supply. When the BOS is activated, all ACBG shall be provided solely by the BOS (no mixing with OBOGS gas) to meet the requirements stated in Table I.

b. The BOS shall not back-flow into the OBOGS plenum.

c. The system shall be capable of manual activation and deactivation by each aircrew member.

d. An indication of BOS activation and the quantity of oxygen remaining in the BOS shall be readily visible to the aircrew with one or more of the following: pressure gage, quantity gauge, electronic display.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

e. The backup oxygen shall flow until the primary breathing gas system (PBGS) achieves a stable and physiologically acceptable pressure and oxygen concentration.

f. In the case where an OBOGS is used as the PBGS, the following failure modes shall apply for BOS activation:

(1) OBOGS inlet supply pressure that is below the minimum inlet air pressure specified for proper operation of the concentrator.

(2) OBOGS product pressure that is below the minimum specified for proper operation of ACBS components such as oxygen monitor and breathing regulators.

(3) Low product oxygen concentration (O_2 concentration below caution line, see caution line shown on Figure 2b).

(4) Critical faults as determined by the PBGS.

g. BOS volumetric sizing should plan for aircrew minute volume consumption rates of 35 LPM (ATPD) per aircrew member for a duration determined by the program office to meet mission requirements during emergency conditions.

5.4.1.2 Emergency oxygen system (EOS).

The EOS is used to provide breathing gas in the event that both the PGBS and BOS fail.

a. When the EOS is activated, all ACBG shall be provided solely by the EOS (no mixing with OBOGS gas) to meet the requirements stated in Tables I, II, and III.

b. EOS shall provide ≥94% oxygen, be auto-activated in no longer than two (2) seconds upon cabin decompression, and be delivered to each aircrew member's mask when no more than 0.6 L of breathing gas has been inspired. The EOS shall also be able to be manually activated and deactivated by each aircrew member.

c. The EOS shall provide the quantity of breathing gas as required to support missions and conditions as specified by the Aircraft Program Office. Consideration should be given to aircraft mission and requirements to satisfy a "get me down" or "complete the mission" based on operational risks.

(1) Rapid Decompression. As a "get me down" system following rapid decompression, the EOS should allow aircrew to safely descend from all altitudes the aircraft is designed to fly to an altitude below 25,000 feet MSL where the EOS is no longer required, assuming that the PBGS is nominally functioning.

(2) PBGS and BOS Failure. If both the PBGS and BOS fail, the EOS shall allow aircrew to safely descend from the aircraft design ceiling altitude to an altitude of 10,000 feet MSL, where the EOS would no longer be required.

d. EOS sizing shall be calculated using aircrew minute ventilation of 35 LPM (ATPD) per aircrew member.

e. An indication of the quantity and/or pressure of oxygen remaining in the EOS shall be readily visible to the aircrew. This information shall be available for electronic transmission to the aircraft mission computer.

f. The EOS shall be capable of manual activation and deactivation by each aircrew member.

5.4.1.3 Ejection Oxygen System.

During ejection, the Ejection Oxygen System shall fulfill the following requirements:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

a. When the Ejection Oxygen System is activated, all ACBG shall be provided solely by the Ejection Oxygen System (no mixing with OBOGS gas) to meet the requirements stated in Tables I, II, and III.

b. The Ejection Oxygen System supply used to provide breathing gas in the event of an ejection from the aircraft, and shall provide ≥94%.

c. Shall be auto-activated at the time of ejection.

d. Shall be capable of manual activation by the aircrew.

e. Shall provide the aircrew sufficient breathing gas volume to allow for post ejection descent from all altitudes the aircraft is designed to fly to a physiologically safe altitude based on aircrew consumption rates of at least 35 LPM (ATPD). Refer to AGARD-AG-286, for descent times.

f. May be combined with the EOS as long as there is sufficient quantity to meet in-aircraft emergencies as well as a reserve for ejection. The aircrew must be notified when the minimum quantity for safe ejection is reached.

g. The ejection oxygen supply quantity (or pressure) shall be visible for pre-flight and in-flight inspection by each aircrew member.

5.5 ACBS Integration.

5.5.1 Backflow prevention.

If the ACBS includes OBOGS, there shall be no loss of product gas backwards through the OBOG. Further, there shall be no reverse airflow at the OBOG inlet.

5.5.2 Breathing gas quality.

5.5.2.1 ACBS requirements.

a. The breathing gas produced by the ACBS shall have at least 95% filtration efficiency for 0.1 to 0.3 µm particles and 99.5% or higher for ≥0.75 µm particles.

b. The ACBS shall meet the CBRN requirements as established by the Aircraft Program Office.

c. During normal operations, the breathing gas shall have no objectionable odor. The breathing gas should not cause aircrew breathing discomfort in nasal passages, throat, mouth, and lungs, etc.

5.5.2.2 OBOGS used as PBGS.

For aircraft that utilize an OBOGS for the PBGS, the following requirements shall apply.

a. The ACBS shall provide breathing gas that meets specifications listed in Table IV, “Quantity at Outlet” column. Additional detail regarding breathing gas quality can be found in Appendix B. The Aircraft Program Office should be contacted if any of the results are exceeded. For contaminants not listed in Table IV that may be discovered by the contractor, the contractor shall identify and report the contaminants to the Aircraft Program Office. The Program Office can contact 711th HPW or NAVAIR, 711.hpw.rh.workflow@us.af.mil, or NAWCAD_HSD_NAWCAD_HSED_MAIL@navy.mil for assistance.

b. The OBOGS supply air shall contain an inlet particulate filter capable of capturing particles of a size of 1 micron or greater.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

c. The OBOGS shall incorporate a system to remove liquid water from air source to prevent degradation of the OBOGS performance and reliability. The design should consider cold weather operations and the possibility of freezing.

   (1) The OBOGS inlet air connection should not be located at a low point in the air supply piping.

d. The ACBS and air source(s) shall implement designs to mitigate aircraft engine exhaust gas ingestion during ground or shipboard operations, and fuel spillage during aerial refueling.

e. The ACBS and air source(s) design shall prevent failure modes where aircraft fluids (hydraulic fluid, coolant fluid, etc.) leak into the ACBS source air.

f. For OBOGS using molecular sieve technology, the molecular sieve pellets shall be secured within the canister to prevent movement and breakdown due to pressurization cycles and vibration. Additionally an outlet filter shall be used to prevent release of molecular sieve particles into the breathing gas.

TABLE IV. ACBS maximum allowable contaminant concentrations in breathing gas.

SUBSTANCE	MAXIMUM ALLOWABLE CONTENT at INLET (ACBS source air respective to airframe)	MAXIMUM ALLOWABLE QUANTITY at OUTLET[1/]
Acrolein	0.1 ppmv	0.05 ppmv
Aldehydes	1 ppmv	0.2 ppmv
Aromatics	10[1/] ppmv	0.1 ppmv
Carbon Dioxide	5000 ppmv	500 ppmv
Carbon Monoxide	50 ppmv/250[2/] ppmv	10 ppmv
Cobalt	0.1 mg/m^3	0.025 mg/m^3
Ethanol	1000 ppmv	500 ppmv
Fluorine (as HF)	0.1 ppmv	0.05 ppmv
Halogenated Solvents	2* ppmv	0.2 ppmv
Hydrogen Peroxide	1 ppmv	0.5 ppmv
Methyl Alcohol	200 ppmv	100 ppmv
Methyl Bromide	20 ppmv	1 ppmv
Nickel	0.5 mg/m^3	0.125 mg/m^3
Nitrogen Oxides	5 ppmv	0.1 ppmv
Oil Breakdown Product	1 ppmv	0.1 ppmv
Oil and Particulate Matter	2[1/] mg/m^3	0.2 mg/m^3

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

TABLE V. ACBS maximum allowable contaminant concentrations in breathing gas - Continued.

SUBSTANCE	MAXIMUM ALLOWABLE CONTENT at INLET (ACBS source air respective to airframe)	MAXIMUM ALLOWABLE QUANTITY at OUTLET1/
Ozone	0.1 ppmv	0.05 ppmv
Sub-Micron Particles	0.5 mg/m^3	0.05 mg/m^3
Total Hydrocarbons	250[1/] ppmv	25 ppmv
Unsaturated Hydrocarbons (alkenes, alkynes)	2[1/] ppmv	0.2 ppmv
Vapor Phase Water	≤95% Non-Condensing	-4 °F Dew Point
NOTES: 1/ The maximum allowable quantity shall be less than the listed quantity. 2/ Current Navy (aircraft carrier) requirement – may apply in other close proximity operations for other aircraft/all Services.		

5.5.3 Temperature.

a. The breathing gas temperature at the aircrew mask during normal ground and flight operations shall be no greater than 10 °F above and no less than 20 °F below the cabin temperature.
b. Oxygen storage vessels shall not be located near other equipment that produces excessive heat loads that may result in a hazardous condition (e.g. fire/explosion, material fatigue, oxygen rich environment).

5.5.4 ACBS monitoring, warning, and recording.

a. At a minimum, the parameters shown in Table V shall be (except for parameters marked as “desired”) measured and recorded as part of the aircraft integrity data and crash survivable recorder. The intent is to have ACBS operational data available in real time, as well as downloadable, to provide inputs for caution and warning alerts and to support post-flight maintenance. Ultimately, these data can be utilized for real-time performance and/or health monitoring and feedbacks for implementing automation control, etc., and adaptive control for future health monitoring capability. If data is stored within the component (e.g., the concentrator), the last 24 hours of operation shall be recorded, at a minimum.

b. Recorded data shall be unclassified and downloadable in a format that can be viewable at field locations by Service personnel. The intent is to have ACBS operational data available in real time, as well as downloadable, to provide inputs for caution and warning alerts and to support post-flight maintenance.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

TABLE VI. ACBS performance data recording.

Parameter	OBOGS	GOX/LOX	Minimum Record Rate (Hz)
Time	Aircraft time	Aircraft time	10
Supply Status - LOX - GOX - BOS - BOS/EOS Combo	 N/A N/A Quantity or Pressure Quantity or Pressure	 Quantity Quantity or Pressure Quantity or Pressure Quantity or Pressure	 1 1 1 1
Concentrator Inlet	Pressure Temperature Humidity/Water[1/]	N/A N/A N/A	10 10 2
Concentrator Outlet	Pressure Oxygen Concentration Flow[1/]	N/A N/A N/A	10 10 10
Breathing Regulator Inlet	Pressure Oxygen Concentration	Pressure N/A	10 10
Breathing Regulator Outlet	Pressure[1/] Flow[1/]	Pressure[1/] Flow[1/]	10 10
Mask	Pressure[1/] Flow[1/] Oxygen Concentration[1/] CO_2 Concentration[1/]	Pressure[1/] Flow[1/] Oxygen Concentration[1/] CO_2 Concentration[1/]	20 20 20 20
Aircraft and Cabin Pressures	Aircraft Pressure Altitude Cabin Pressure	Aircraft Pressure Altitude Cabin Pressure	10 10
Operation	System Faults	System Faults	10
Auxiliary Systems	Activation Enunciation	Activation Enunciation	2
Warnings	Low Inlet Pressure Low Outlet Pressure Low O_2 Concentration Other[2/]	Low Quantity Low Pressure N/A Other[2/]	2 2 2 TBD

NOTES:

1/ Desired.

2/ Any other warning that would compromise human performance or indicate ACBS specification not being met.

Breathing regulator pressure and mask pressure should be measured relative to the cabin.

Oxygen concentration is expressed in percent by volume or as oxygen partial pressure.

5.5.5 Advisories, cautions, and warnings.

The pilot shall be notified of any faults that might compromise human performance. The advisory, warning, or caution indications in Table VI should be enunciated to the aircrew and recorded in the aircraft integrity data and the aircraft Crash Survivable Memory Unit (CSMU), if the aircraft is

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

equipped with a CSMU. Advisory/Caution/Warning message requirements should be tailored for the aircraft application and should consider the appropriateness of the indicator as well as continuity of training for the aircrew. Persistence requirements shall be tailored and approved by the program office. (Persistence is the duration that caution/warning is active before it is annunciated to the aircrew.)

a. An oxygen caution shall be activated at and below the calculated caution line (see Figures 2a and 2b).

b. An oxygen warning shall be activated at and below the red warning line (see Figures 2a and 2b).

c. An advisory shall be provided to maintenance personnel post flight if any oxygen concentration deviation outside the design zone, as shown on Figures 2a and 2b, occurs during flight.

TABLE VII. Advisory, caution, and warning indications.

Indication	Level
BOS ON	Advisory
LO SUP PRESS – OBOGS (Low inlet pressure)	Advisory
BOS LOW (<10% volume remaining)	Caution
BOS FAIL	Caution
EOS ON	Caution
PBGS FAIL (warning line, see Figure 2b)	Caution
OXY LOW – LOX/GOX (below minimum quantity/pressure)	Caution
OXY LOW – OBOGS (calculated caution line, see Figure 2b)	Caution
OXY FAIL (PBGS AND BOS FAIL)	Warning
EOS LOW (such as minimum safe ejection quantity remaining)	Warning
EOS FAIL	Warning

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

5.5.6 Built-in test functionality (BIT).

The ACBS shall have built-in fault detection and isolation during ACBS start-up and during ACBS operation (in-flight). The percent of faults detected and reporting time shall be determined by the Program Office. BIT data shall be recorded. The BIT shall check the following:

a. Oxygen monitor.

b. Electronic circuits.

c. Pressure regulation, including any pressure reducing valves.

d. Mechanical performance of the oxygen concentrator, including air control valve(s) to verify they are functioning correctly.

e. Startup (power-up) BIT.

(1) BIT shall occur automatically, initiate upon application of power and availability of source air.

(2) The ACBS shall be capable of supporting aircrew breathing by meeting test conditions 1 and 2 (see Table I) within 10 seconds of BIT initiation.

(3) The startup BIT shall be completed within three (3) minutes of BIT initiation: four (4) minutes for extreme cold conditions (< -40 °F).

(4) Startup BIT shall not interfere with aircraft ground operations.

(5) Startup BIT shall be inhibited if the aircraft is airborne.

f. In-flight (continuous built-in test).

(1) The BIT shall not degrade ACBS performance during flight.

g. Maintenance BIT.

(1) The maintainer shall be able to initiate a built-in fault detection test with the aircraft on the ground.

(2) If the ACBS includes OBOG, then BIT shall also test molecular sieve bed health.

5.5.7 Materials and components.

The contractor shall verify that system use does not cause known health risks. This information shall be submitted to the Aircraft Program Office for review and approval.

a. The contractor shall disclose all chemical compounds and Safety Data Sheets as required by the Hazard Communication Standard (see 29CFR 1910.1200). If the system uses an OBOG, the molecular sieve bed adsorbent materials shall be identified.

b. The contractor shall provide evidence that the materials as used in the ACBS are safe for human use in accordance with current U.S. Government occupational health and toxicology standards.

c. The components compatibility with high purity and/or high-pressure oxygen shall be analyzed for potential fire hazards in accordance with ASTM MNL36.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

5.5.8 Human factors.

The ACBS design shall facilitate safe and effective operation of the aircraft while readily providing the aircrew essential information about the ACBS status. The following should be included in the crew station design:

a. The following shall be readily visible (within the forward plane) to the aircrew member: a breathing gas flow indicator; all advisory, caution and warning indications: and all stored oxygen pressures/quantities.

b. An oxygen mask functional test (such as press-to-test) and PBG test (if equipped) at pressures appropriate for the aircraft to show functionality and not harm the aircrew. The mask fit check and PBG test processes shall be coordinated with the Aircraft Program Office and Service Aeromedical SMEs.

c. The ACBS shall allow the aircrew members to perform all of their duties without interference from ACBS sensors.

5.5.9 High pressure oxygen storage.

Storage of high pressure oxygen (above 500 psi) in combat aircraft shall be in cylinders that conform to MIL-DTL-7905. Deviations from MIL-DTL-7905 requires Aircraft Program Office approval.

Storage of high pressure oxygen (above 500 psi) in non-combat aircraft shall be coordinated with the Aircraft Program Office.

5.5.10 Oxygen concentrator waste gas venting.

a. The OBOG exhaust gas shall be sufficiently vented to ensure negligible pressure drop (back pressure) with respect to aircraft barometric pressure which could result in concentrator performance degradations. If vented internally within the aircraft, waste gas shall not be allowed to enter the aircraft cabin.

b. Any waste gas shall be prevented from back flowing into the ECS or bleed air system during low pressure or loss of pressure conditions.

c. Water vented into the waste gas shall be controlled to prevent damage to aircraft structures or avionics.

5.5.11 Validation and verification of ACBS.

The ACBS shall be tested to ensure that all requirements specified in section 5 and those found in Appendices C and D have been met. The ACBS shall also be tested to ensure that any additional requirements as designated by the Aircraft Program Office have also been met.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use.

This standard is intended to provide a minimum set of requirements to ensure the safety and effectiveness of ACBS for military and U.S. Government aircraft.

6.2 Acquisition requirements.

Acquisition documents should specify the following:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

a. Title, number, and date of this standard.

6.3 Patent notice.

The Government has a royalty-free license under the following listed patents for the benefit of manufacturers of the item either for the Government or for use in equipment to be delivered to the Government:

U.S. patent number
US4916630

6.4 Subject term (key word) listing.

Aircrew oxygen
Aviator's breathing oxygen
Backup oxygen system
Breathing regulator
Breathing resistance
Contaminants
Ejection oxygen system
Emergency oxygen system
Hypercapnia
Hyperoxia
Hypocapnia
Hypoxia
Life Support
LOX
Mask pressure
Molecular sieve
On board oxygen generating system
Oxygen
Oxygen concentration
Oxygen mask
Oxygen partial pressure
Oxygen system
Pressure breathing
Safety pressure
Work of breathing

6.5 International standardization agreement implementation.

This standard implements AIR STD AMD 4068 Ed 2 v2, Physiological Requirements for Aircrew Oxygen Masks for Use at High Breathing Pressures. When changes to, revision, or cancellation of this standard are proposed, the preparing activity must coordinate the action with the U.S. National Point of Contact for the international standardization agreement, as identified in the ASSIST database at https://assist.dla.mil.

6.6. Changes from previous issue.

Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

APPENDIX A

PHYSIOLOGICAL REFERENCE INFORMATION AND CALCULATIONS OF RESPIRATORY LOADS IMPOSED BY A BREATHING SYSTEM

A.1 SCOPE

A.1.1 Background.

This appendix provides guidance on how to determine relevant pressures and on how to measure work of breathing of an ACBS. The values obtained can be used to judge acceptability of an ACBS.

An ACBS imposes loads on the respiratory muscles. Typically, breathing resistance, the only load quantified by work of breathing measurements, is the biggest concern. However, respiratory loads are also imposed by the elastic stretch of flexible hoses and masks, static pressures (e.g. safety pressure) and by pressure thresholds to start flow. In addition, a low flow ACBS may restrict the volume available for each breath. Inhalation and exhalation loads must be judged separately since they affect different muscle groups.

This Appendix is not a mandatory part of the standard. The information contained herein is intended for guidance only.

A.1.2 Perspective.

Historically, peak pressures have been used to judge whether a breathing apparatus imposes an acceptable breathing resistance, but those measurements fail to capture the durations of the extreme pressures. The volume-weighted average pressure may better reflect total effort, but that measurement may miss the importance of a large, short-lasting pressure spike. A better way to judge the respiratory load is to use both peak and volume-weighted mean pressures.

The volume-weighted mean pressure is calculated as work of breathing divided by the tidal volume (V_T), the volume of the breath. Like the assessment of peak pressure as a function of peak flow in the ranges expected in aircraft operations, work of breathing should be assessed as a function of minute ventilation (minute volume) across the expected values.

The respiratory physiology methods for measuring the work of breathing were already well developed in the 1950s. These methods have since been adopted in other fields of respiratory protection to determine the work imposed by a breathing apparatus in a standardized and repeatable way.

A.2 ACBS MINUTE VENTILATION AND PEAK FLOWS

The ACBS should provide sufficient flow to meet aircrew breathing demands for the conditions the aircraft is designed to operate. Typical minute ventilations for various flight conditions are shown in Table A-I. The ACBS should be capable of achieving peak inhalation and exhalation flows as described in Table A-II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

TABLE A- I. Typical ACBS minute ventilation.

Flight Condition	Typical minute ventilation (CLPM) at all altitudes (per crew member)
Ground Operations	22
Take-Off	25
Take-Off (Carrier Based)	28
Landing (Conventional)	28
Landing (Carrier Based, Vertical)	35
Idle Descent with Low Workload	25
Moderate Workload	35
Routine In-Flight, Navigation	25
Post G recovery	50
Dynamic Maneuvering (up to 5 Gz)1/	42
Dynamic Maneuvering (up to 7.5 Gz)1/	53
Dynamic Maneuvering (up to 9 Gz)1/	35
Passengers	16
1/ for 20 second duration	

TABLE A- II. Instantaneous peak inhalation and exhalation flows for sinusoidal breathing (liters ambient temperature and pressure dry (ATPD) per person).

Flight Condition	Peak Flow (CLPM)
Ground Operations	85
Take-Off	105
Take-Off (Carrier Based)	125
Landing (Conventional)	123
Landing (Carrier Based, Vertical)1	110
Idle Descent	115
Moderate Workload	110
Routine In-Flight, Navigation	120
Post G recovery	160
Dynamic Maneuvering (up to 5 Gz)	180
Dynamic Maneuvering (up to 7.5 Gz)	258
Dynamic Maneuvering (up to 9 Gz)	322

A.3 IDENTIFICATION OF PEAK PRESSURES

A.3.1 Peak pressures.

The pressure variations in the crewmember interface can be plotted against time. Figure A-1 shows pressure variations for an ACBS for three stylized sinusoidal breaths, each starting with

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

inhalation, when breathed on by a person or by a breathing machine. The leftmost, black line shows pressures without safety pressure; the middle, grey interrupted line shows pressures for a partial safety pressure system, and the rightmost, blue line shows a system with safety pressure. These three modes are defined more fully below. The peak pressures are shown as $\Delta p_{in,\ min}$ and $\Delta p_{ex,\ max}$, where "Δ" means difference from cabin pressure, "in" refers to Inhalation, "ex" means exhalation, "min" is the lowest (minimum) pressure and "max" is the highest (maximum) pressure.

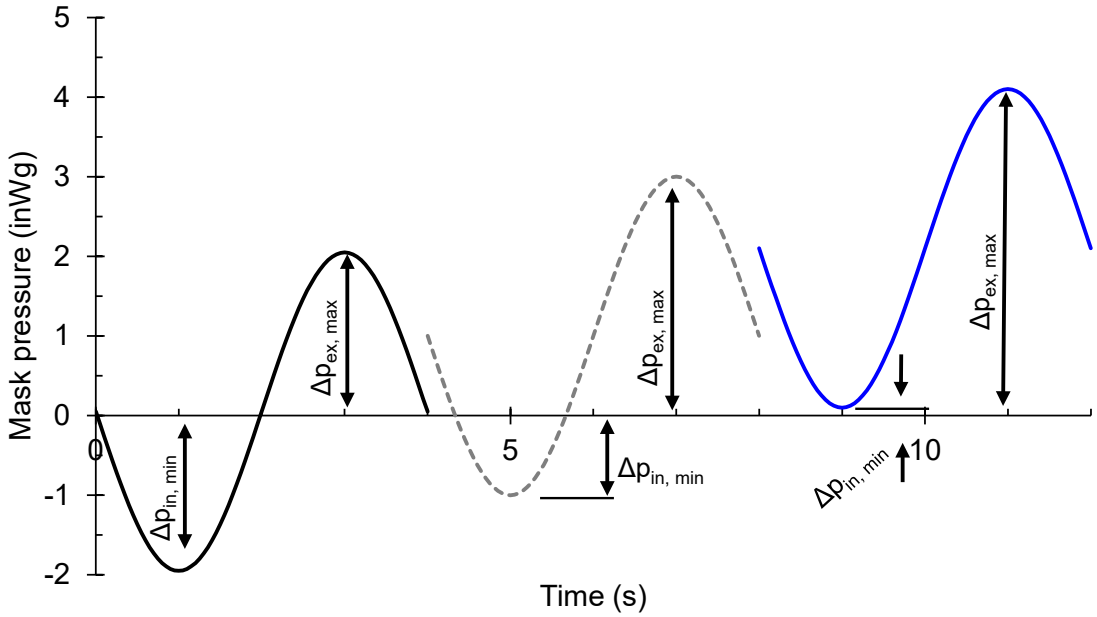


FIGURE A-1. Peak pressure identification.

A.3.1.1 No safety pressure.

In a system without safety pressure, the pressure at the crew interface has to go negative (below cabin pressure) for all inhalation gas. This system provides the least protection against inward inhalation (leaks) of cabin air, which might be hypoxic or contaminated.

A.3.1.2 Partial safety pressure.

A system with partial safety pressure permits some inhalation flow when the pressure at crew interface is above cabin pressure, but allows the pressure at the crew interface to go negative for some gas flows. Typically, negative crew interface pressures may begin partway through inspiration and in the higher flow ranges. Inward leaks could occur with high inhalation flow.

A.3.1.3 Continuous safety pressure.

A system with continuous safety pressure does not allow the pressure at the crew interface to go negative for any gas flow within its intended range. This type of system provides the highest protection against inward leaks.

A.3.2 Identification of peak pressures and work of breathing.

Figure A-2 shows how the pressure and volume can be plotted together to determine work of breathing graphically. In this example, gas is breathed from a source at constant pressure equal to the cabin pressure, and exhaled to cabin pressure. The pressure axis thus represents the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

pressure in the aircrew interface relative to cabin pressure. Inhalation pressure swings are negative.

A.3.2.1 Pressure-volume loop (ACBS with no safety pressure).

Figure A-2 (left plot) shows how inhalations follow the curve ACB and exhalations follow the curve BDA. Points A and B indicate the transitions between inhalation and exhalation when flow is zero. The volume difference between points B and A is the tidal volume. The pressure at point C is the peak inhalation pressure ($\Delta p_{in,\ max}$) and that at point D is the peak exhalation pressure ($\Delta p_{ex,\ max}$). Peak pressures are functions of peak flow and resistance to flow. The pressure difference between points D and C is the peak pressure swing. An increase in minute ventilation requires a larger tidal volume, higher peak flow, or both. Thus, pressure-volume loops like that shown will contain a larger area for higher, relative to lower, minute ventilation.

For an ACBS that has only resistive characteristics, the pressures at the no-flow points A and B will be the same. Any elasticity in the ACBS (e.g., flexible or pumping hoses, or a flexing mask) will be indicated by a lower pressure at point B (end inhalation, start exhalation) than at point A (end exhalation, start inhalation). The pressure difference between points A and B is the pressure required to distend the elastic components.

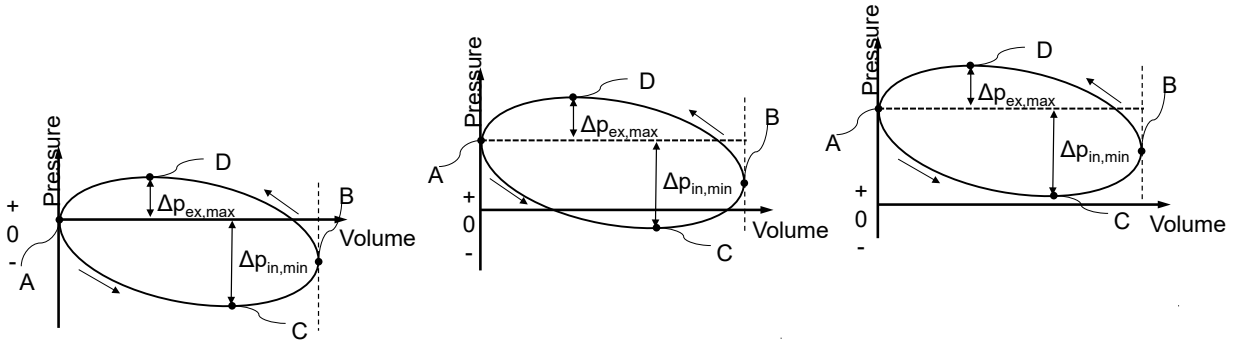


FIGURE A-2. Pressure-volume diagram of a breath from an ACBS. No safety pressure (left), partial safety pressure (center) and full safety pressure (right).

A.3.2.2 Work of breathing (WOB) and average pressure.

The area enclosed by the pressure – volume loop, i.e., pressure change multiplied by volume change (equivalent to force multiplied by displacement) is the resistive WOB, i.e., the work to move the gas into and out of the aircrew interface against the flow resistances of the ACBS. The total WOB can be separated into the inhalation and exhalation components. The inhalation WOB is the area between the curve ACB and a line drawn between points A and B (area inside ACBA). The exhalation WOB is the area between the curve BDA and a line drawn between points B and A (BDAB).

Mathematically, WOB for any breath is the integral of pressure by increments of volume, $\int P \cdot dV$ or $\Sigma(P \cdot \Delta V)$ for digitized signals.

The WOB is typically reported as the WOB divided by the tidal volume to yield the volume-weighted averaged pressure. The SI units are J/L = kPa.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

A.3.2.3 Magnitude of elasticity.

The magnitude of elasticity can be calculated from the pressure difference between points B and A, divided by the tidal volume. Elasticity has the units of pressure/volume.

A.3.2.4 Pressure-volume loop (ACBS with partial safety pressure).

Figure A-2 (center plot) illustrates the pressure-volume loop for an ACBS that provides partial safety pressure. The primary difference from the Figures A-2 and A-3 is that the pressure in the mask goes negative partway into the inhalation. The pressure of zero is the cabin pressure. Calculations are performed as above (see A.3.2.2).

A.3.2.5 Pressure-volume loop (ACBS with safety pressure).

Figure A-2 (right plot) illustrates the pressure-volume loop for an ACBS that provides safety pressure. The primary differences from an ACBS without safety pressure (see Figure A-2) are (1) that the loop has been shifted upwards and that (2) point C (the lowest pressure in the mask) is positive relative to cabin pressure. The interrupted constant-pressure line starting at point A shows the safety pressure which is a pressure offset. The pressure of zero is the cabin pressure. Calculations are performed as above (see A.3.2.2).

A.4 ADDING UP THE RESPIRATORY LOADS

A.4.1 Total respiratory load.

The total respiratory load imposed on either the inhalation or the exhalation muscles can be determined as the sum of individual loads if they are expressed as percentages of tolerable load. This way of adding up the total load allows a manufacturer to trade one load for another while still keeping the total load within acceptable limits. For instance, an unavoidable load can be compensated for by keeping another load low. The method also allows a testing laboratory to extract each load and then calculate the total respiratory load.

Figure A-3 illustrates how the total load can be determined. Assume that the following parameters have been determined at a minute ventilation of 20 L/min: the inhalation and exhalation volume averaged pressures were 0.5 and 0.4 inWg, respectively. The safety pressure was 1.5 inWg (the leftmost point in the graph). The elastance was 0.3 inWg/L (the slope of the interrupted line).

The mean inhalation pressure was greater than the mean exhalation pressure so it was the limiting pressure of the two. Table I shows that, at 20 L/min, the maximum mean pressure is 1.5 inWg. Therefore, the resistive load represented 0.5/1.5 = 33% of its maximum.

The maximum elastic load is 4.0 inWg/L (see Table I). The elastic load was 0.3/4.0 = 7.5% of its maximum.

The maximum static pressure is 4 inWg (see Table I). The worst-case safety pressure load was 1.5/4 = 37.5%.

The total load becomes 33% + 7.5% + 37.5% = 78% of maximum.

The remaining 22% could allow a designer to adjust one or more of the loads. However, it should be kept in mind that a lower total load is better for the aircrew.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

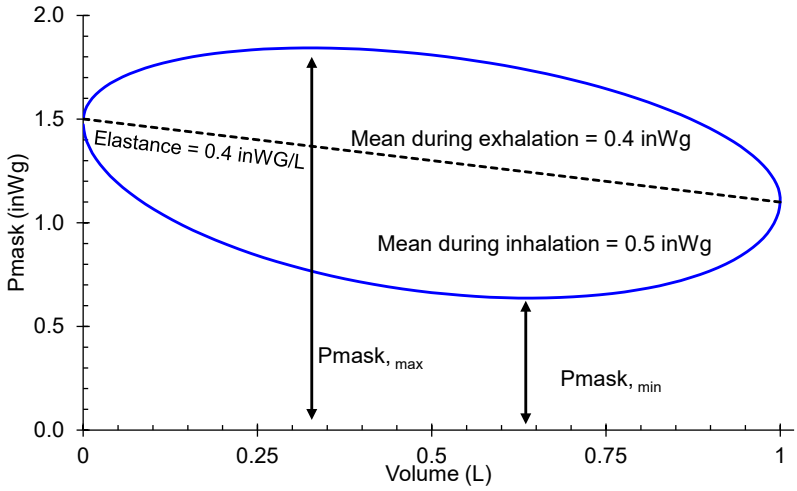


FIGURE A-3. Pressure-volume diagram with example values.

A.5 PROCEDURE

A.5.1 Parameters to be reported.

For any mode of safety pressure operation, plot the pressure on the y-axis against lung displacement volume on the x-axis. Report the peak pressures (points C and D on the Figures above). Numerically determine the areas for inhalation and exhalation work of breathing, and calculate the volume-weighted mean inhalation and exhalation pressures, that is, work of breathing divided by tidal volume. Report the mean values separately for inhalations and for exhalations. The pressure at point A (just before inhalation flow begins) should be reported as the safety pressure. Determine and report the peak pressures. Calculate and report the elastance and the total respiratory loads.

REFERENCES

NATIONAL LIBRARY OF MEDICINE (NIH)
"Physiologically Acceptable Resistance of an Air Purifying Respirator". Ergonomics. 2011; 54(12):1186-1196; Barbara E. Shykoff, Dan E. Warkander.

(Copies of this document are available online at https://pubmed.ncbi.nlm.nih.gov/22103726/.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

APPENDIX B
ACBS INTEGRATION READINESS

B.1 SCOPE

B.1.1 Scope.

This appendix details the approach for ACBS integration readiness. Integration readiness is a critical process for ensuring a safe and effective ACBS. The Aircraft Program Office shall oversee the coordination of this process. This Appendix is a mandatory part of the standard. The information contained herein is intended for compliance.

B.2 MEETINGS

B.2.1 Design/Redesign meetings.

The Aircraft Program Office shall conduct periodic meetings to include the ACBS contractor, the aircraft prime contractor, Aircrew Flight Equipment personnel, Life Support design teams, and Aerospace Physiology technical experts during the ACBS system development/modification process to ensure comprehensive consideration of ACBS integration issues. The contractor shall inform the program office of any safety-of-flight issues that arise following installation or operational use/maintenance of ACBS products. The contractor shall also be called to present at any meeting regarding any safety-of-flight issues that are identified with the contractor's product. If the ACBS includes OBOGS, then the ACBS contractor shall coordinate with the Aircrew Flight Equipment, Environmental Control System, and Engine design teams during the development process to evaluate the Oxygen Generator and its air source (i.e., environmental control system and engine bleed air) interfaces.

B.2.2 Evaluation criteria.

As a minimum, the meetings shall evaluate the interface criteria listed below:

a. ACBS minimum in-flight inlet air pressure and conformance to the ACBS specification.

b. ACBS minimum ground inlet air pressure and conformance to the ACBS specification.

c. ACBS maximum inlet air temperature and conformance to the ACBS specification.

d. ACBS inlet air pressure while other aircraft subsystems are demanding airflow at their maximum rate and conformance to the ACBS specification.

e. Engine bleed air quality and conformance. Engine generated substances contained in the bleed air shall be within the threshold limit values specified within this standard (see Table IV, second column). Compliance with specified threshold limits for the listed substances shall be demonstrated via analysis of bleed air samples obtained during live engine runs. If substances other than those listed are present in the air samples, the engine manufacturer shall report the substance(s) and the contamination in parts-per-million to the Aircraft Program Office for determination of maximum limits. When two or more engine-generated substances are present, their combined effect shall be determined and reported. In the absence of information to the contrary, the combined effects of the different substances shall be considered as additive.

    (1) If cleaning fluids are specified for use during normal engine maintenance, then engine bleed air samples shall be tested during the development and test of the cleaning process to verify air quality meets the limits in Table IV, second column,

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

and requirements in 5.5.7. Any contaminants not listed in Table IV that are discovered shall be identified by the contractor and reported to the Aircraft Program Office.

f. Predicted frequency and potential causes for in-flight loss of ACBS inlet air pressure.

g. Predicted frequency and potential causes for in-flight ACBS inlet air low pressure transients (i.e., below the ACBS minimum inlet air pressure specification).

h. Predicted frequency and potential causes for in-flight deviations of mask pressure, mask flow, and mask/crewmember interface oxygen concentrations.

i. Predicted frequency and potential causes for in-flight failures of either or both the regulator and the mask valve, resulting in deviations of mask pressure, mask flow, and mask/crewmember interface oxygen concentrations. This analysis shall be conducted in light of any relevant physiological episode findings.

B.3 REVIEWS

B.3.1 Engineering technical reviews.

For new aircraft acquisitions, the Aircraft Program Office shall review all documentation supporting ACBS integration readiness at the Aircrew Flight Equipment, Environmental Control System, and Engine Preliminary Design Reviews (PDRs) and Critical Design Reviews (CDRs). ACBS integration readiness shall be reviewed during Aircrew Flight Equipment, Environmental Control System, and Engine technical reviews. At a minimum, the interface characteristics noted in B.2.2 shall be reviewed. Any conditions causing an ACBS specification non-compliance shall be identified and analyzed. Approaches for mitigating any out-of-specification condition shall be evaluated. Engine bleed air and ECS modeling results or data shall be used to determine that the oxygen generator inlet pressure and temperature interface conditions will be met throughout ground and flight operations. An ACBS Integration Readiness Document summarizing the integration readiness analysis, results, and conclusions shall be reviewed and approved by the Aircraft Program Office.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

APPENDIX C
CONTRACTOR VERIFICATION AND VALIDATION OF ACBS

C.1 SCOPE

C.1.1 Compliance.

This appendix details the methods of compliance and substantiating data of compliance that apply to ACBS. The intent is to identify a minimum set of required and suggested tests. Government Verification/Validation is addressed in Appendix D. This appendix is a mandatory part of the standard. The information contained herein is intended for compliance.

C.1.2 Verification testing.

ACBS verification testing shall be completed as directed by the Aircraft Program Office. The Aircraft Program Office should coordinate with subject matter experts from the AFRL and Navy Aircrew Systems Program Office (SPO).

C.2 APPLICABLE DOCUMENTS

C.2.1 Government documents.

C.2.1.1 Specifications, standards, and handbooks.

The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-810	Environmental Engineering Considerations and Laboratory Tests

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-516	Airworthiness Certification Criteria

(Copies of these documents are available online at https://quicksearch.dla.mil.)

C.2.2 Non-Government publications.

The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM MNL36	Safe Use of Oxygen and Oxygen Systems: Handbook for Design, Operation, and Maintenance

(Copies of this document are available online at www.astm.org.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

C.3 METHODS OF COMPLIANCE

C.3.1 System hazard and failure analysis.

The contractor shall conduct an oxygen system safety analysis of the compatibility of the life support components, including ACBS, to ensure the components are compatible with oxygen at the pressures and temperatures used. The system safety analysis shall include an oxygen compatibility analysis (fire hazard analysis) in accordance with ASTM MNL36 and failure mode effects criticality analysis.

C.3.2 Qualification.

The contractor shall qualify the ACBS components, including environmental, acceleration, vibration, and electromagnetic interference testing. The testing shall comply with MIL-STD-810, but may be tailored to the specific aircraft operating conditions if approved by the Aircraft Program Office. The system components shall pass qualification testing prior to initiation of safe-to-fly testing. ACBS testing shall be conducted in order to validate that it meets human physiological requirements and is required as part of any new system/aircraft design and upgrade acceptance or modification of ACBS. ACBS testing shall be conducted during monitored acceptance flight and regularly scheduled maintenance/flight control flights.

For OBOGS, the contractor shall conduct qualification testing of breathing gas quality, including maximum allowable contaminant concentrations. The Aircraft Program Office will provide test and verification methods.

The contractor shall conduct qualification testing utilizing representative operational profiles and breathing demands provided by the Aircraft Program Office.

C.3.3 Airworthiness certification.

The program office shall demonstrate compliance with MIL-HDBK-516.

C.3.4 Aircraft delivery.

The contractor shall verify prior to Government acceptance of an aircraft that the ACBS meets all requirements established by the Aircraft Program Office and functions appropriately during a functional check flight.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

APPENDIX D

GOVERNMENT VERIFICATION AND VALIDATION OF ACBS

D.1 SCOPE

This appendix details the methods of compliance and substantiating data of compliance that apply to acquisition of an ACBS and any subsystem thereof. The intent is to identify a minimum set of required and suggested tests that will define the operational environments of the systems, but not every possible test. Aircraft Program Offices may need to add other tests, as necessary. The verification testing shall be conducted in approved test laboratories followed by in-aircraft ground and flight operations while aircrew are utilizing the ACBS. This appendix is a mandatory part of this standard. The information contained herein is intended for compliance.

D.2 APPLICABLE DOCUMENTS

D.2.1 Other Government documents, drawings, and publications.

The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AIR FORCE INSTITUTE OF TECHNOLOGY (AFIT) STAT CENTER OF EXCELLENCE (COE)

https://www.afit.edu/stat

D.3 METHODS OF COMPLIANCE

D.3.1 Digital engineering.

The Aircraft Program Office shall obtain the following from the contractor as soon as these are available:

a. Interface Control Document.
b. A solid digital model Computer Aided Design (CAD) of the entire ACBS.
c. A digital twin of the ACBS system to facilitate digital engineering and virtual modeling (if available).

D.3.2 Scientific Test and Analysis Techniques (STAT).

The Aircraft Program Office shall use scientific test and analysis techniques that will produce the required data to characterize system behavior across an appropriately selected set of factors and conditions. This is intended to apply to the entire ACBS system as well as all subsystems of an applicable crew breathing system. It is critical that the Aircraft Program Office documents the test strategy and shares the strategy and methods with the developing contractor to ensure program objectives for rigor and schedule are met. Careful consideration should be given at every stage for incorporating STAT best practices as identified on the following website: https://www.afit.edu/stat.

STAT describes the scientific and statistical methods that enable the development of efficient rigorous test strategies yielding defensible results. STAT incorporates, but is not limited to, such techniques as design of experiments (DOE), reliability growth, software coverage, and survey design used within a larger decision support framework. The suitability of each method is

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

determined by the specific, measureable, unbiased, and of practical consequence test objectives that assist the PM in understanding and quantifying technical risk.

D.4 QUALIFICATION TESTING

D.4.1 Test preparation.

The Aircraft Program Office shall coordinate with subject matter experts from organizations such as AFRL, NAMRU-D, and Navy Aircrew Systems Program Office to develop the test strategy and test plans.

D.4.2 Test hardware.

a. A sufficient number of ACBS systems and subsystems to ensure rigorous testing.

(1) A complete *aircraft representative* ACBS with at least (where applicable):

a. ALL plumbing (pipes, tubes, fittings, air supply line, etc.)

b. Concentrator

c. Plenum

d. BOS

e. Regulators

f. Valves

g. Electrical connectors

h. Wiring bundles

i. Aircraft interface software

(2) Additional components shall be acquired for testing at the extremes of the specifications that may result in degraded or destructive testing.

b. Contractors are free (subject to their contract) to independently develop their own test strategy and methods, but must consider the delivered systems shall be subject to rigorous Government testing in accordance with the military standards and other documents referenced in the main portion of this standard.

D.4.3 ACBS safe-to-fly testing.

The ACBS shall undergo safe-to-fly testing prior to aircraft flight-testing. Safe-to-fly testing shall be conducted by a Government agency. The test agency shall prepare the test plan, including success criteria. Prior to testing, the test agency shall coordinate the test plan with the contractor and the Aircraft Program Office. The Aircraft Program Office should coordinate with subject matter experts from the AFRL and Navy Aircrew Systems Program Office. Test plans should be shared with the developing contractor for consistency. This will ensure that objectives are tracked across the scope of testing and that STAT methods are incorporated along each step.

D.4.3.1 Testing with breathing simulators.

a. If the PBGS system is to be in use on multiple aircraft, the system, subsystems, and components under test shall be tested against the parameters of each aircraft that the ACBS will be used in. These parameters include but are not limited to maximum Gz, the maximum service ceiling, maximum altitude change rates (ascent and descent), etc. It is recommended that at least a 10% safety margin be added to the unoccupied test. Example: if the operational ceiling is 50,000 feet, test to 55,000 feet. The meaning of safe but substantial margin must be defined by the program office.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

b. Unoccupied testing phases shall, at a minimum, measure the breathing gas oxygen concentration, flow, and temperature delivered to the masks and the pressure within the masks simultaneously for a fully-crewed aircraft. All of the above parameters shall be measured and recorded along with the timestamp at least 20 times per second (20 Hz) throughout the testing. Sample rates of at least 100 Hz should be employed for mask pressure and flow measurements when using Table I, test conditions 11-13.

c. The tested altitudes shall cover the operational altitude envelope. Cabin altitude will be set based on the appropriate pressure differentials. Aircraft and cabin altitude shall be tested at incremental levels from ground to operational ceiling. At a minimum, tests should be conducted at pressurized and unpressurized cabin altitudes up to the service ceiling in both directions. Altitude change rates shall cover the full operation range from maximum ascent to maximum descent, as well as for rapid decompression and insidious decompression scenarios.

d. For OBOGS equipped aircraft, the inlet pressure shall be varied across the expected operational range. Additional testing shall use lower pressure settings than the specified minimum to test the persistence of the system during short duration pressure degradation and of the backup and emergency response for longer durations. Durations for low pressure shall cover the range from 5 to 60 seconds. Inlet pressure of zero shall also be tested to determine the effects of total loss of inlet pressure/flow. After exposure to zero and low pressure settings, the inlet pressure shall be returned to nominal until the backup and emergency systems deactivate and the PBGS recovers to characterize the recovery time as a function of low inlet pressure and nominal pressure.

e. Testing shall include failure mode simulations to verify caution and warning recognition and annunciation (see 5.5.5).

f. Testing shall include verification of the digital communication links between the ACBS and mission computer or equivalent.

g. Testing shall include validation that data stored by the ACBS is true and accurate.

h. Testing shall include representative mission profiles provided by the Aircraft Program Office. The Aircraft Program Office should contact Aeromedical and operational SMEs to determine breathing demands and their durations.

D.4.3.2 ACBS chemical contaminant testing (OBOGS equipped aircraft) lab test.

The ACBS shall be tested for contaminants at the levels noted in “Maximum Allowable Content at Inlet” column in Table IV. Testing shall be accomplished at nominal and minimum specification inlet air pressures; minimum and maximum ACBS product gas flow rates; room temperature; and ground level. The ACBS product gas shall not exceed the levels in Table IV, “Maximum Allowable Quantity at Outlet”.

D.4.3.3 Testing with humans.

Testing with humans shall be accomplished if any of the following criteria are met:

a. An aircraft’s ACBS (or a subsystem thereof) is being replaced or significantly modified.

b. During unoccupied testing, pressures at the mask deviate from Figure 3 or Table I when simulating normal or emergency operating conditions.

c. During unoccupied testing anomalies in test data are identified.

d. New or novel technology is being introduced into the ACBS.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

e. The aircraft operating ceiling is above 50,000 feet.

f. If directed by the Aircraft Program Office.

Human testing of the ACBS, if required, shall address the full range of normal cabin altitudes, rapid decompression altitudes, Gz levels, and emergency operating conditions.

D.4.3.4 Safe-to-fly documentation.

The responsible test agency (711th HPW or NAWC HSD) provides safe-to-fly documentation to aircraft program office verifying that it is safe to proceed with in-aircraft testing.

The test agency shall prepare safe-to-fly documentation and final report (if required) discussing the test data/results, pass/fail criteria results, and significant findings. The letter shall state the recommendation for system flight-testing including any restrictions such as maximum ceiling altitude and maximum Gz level, and document any anomalies found.

D.4.3.5 In-aircraft testing.

D.4.3.5.1 Ground testing.

D.4.3.5.1.1 ACBS chemical contaminant testing (OBOGS equipped aircraft) aircraft ground test.

The ACBS and air source(s) (i.e. ECS and secondary air source, if used) shall be instrumented to verify that contaminant concentrations in breathing gas that are listed in Table IV are below the maximum allowance. Gas concentration shall be measured and recorded in the cockpit, as close to the crewmember as practicable. The goal is to ensure the ACBS and the air source(s) maintain acceptable performance under flight conditions and do not violate human physiological requirements, and do not cause an adverse health effect. Samples of the ACBS inlet air and breathing gas shall be collected during ground engine runs. As a minimum, ten (10) samples shall be collected and analyzed from, at a minimum, the ACBS inlet and crewmember interface locations. Any exceedances shall be reported to the Aircraft Program Office.

D.4.3.5.2 Flight testing.

The Aircraft Program Office should collaborate with the Responsible Test Organization and aeromedical SMEs on the ACBS test plan to minimize the risk to the test aircrew.

For new aircraft development or an aircraft's ACBS (or a subsystem thereof) is being replaced or significantly modified, the ACBS (including regulator and aircrew mask) shall be ground and flight tested throughout the environmental extremes that the aircraft is expected to operate such as carrier, desert, tropical, and arctic operations. During the flight test period, ACBS data shall be provided to the Aircraft Program Office. A minimum of 100 flight hours of flight test data using at least two production representative systems (including aircraft) to include data representative of operational environmental extremes shall be analyzed. After the 100 hours of dedicated or piggy-back flight testing, a surveillance program shall be initiated to review ACBS stored flight data on randomly selected flights to assess system performance. For ACBS modification, a tailored flight test program, as directed by the Aircraft Program Office, shall be employed. A qualified Government agency shall conduct flight-testing.

D.4.4 Flight test report.

The flight-testing agency shall prepare a flight test final report. The flight test report shall summarize the results of the ACBS flight-test program, including acquired aircrew physiological data, assess ACBS performance and suitability, and document any anomalies. If anomalies are noted, the flight test agency shall assess the impact on ACBS safety and effectiveness when the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

aircraft is used in its planned mission role. The flight test report should also summarize impact to human physiological requirements.

D.4.5 ACBS CBRN testing.

For aircraft that are designed to operate in a CBRN environment, ACBS verification/validation shall ensure that the ACBS delivers safe breathing gas.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

APPENDIX E

PROCEDURES FOR SUSTAINMENT/MAINTENANCE FOR ACBS

E.1 SCOPE

E.1.1 Scope.

This Appendix details the sustainment/maintenance procedures for ACBS. A fly-to-fail policy shall not be used in lieu of specific procedures (see E.3). The intent is to include this information in the applicable aircraft technical orders. This Appendix is a mandatory part of the standard. The information contained herein is intended for compliance.

E.2 APPLICABLE DOCUMENTS

E.2.1 Other Government documents, drawings, and publications.

The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

U.S. PATENT & TRADEMARK OFFICE (USPTO)

US4916630 Bed Tester for Molecular Sieve Oxygen Concentrator

(Copies of this document are available online at http://www.uspto.gov/.)

E.3 PROCEDURES

E.3.1 Performance check procedures.

Where applicable, the contractor shall incorporate the information below in the aircraft technical orders.

E.3.1.1 Periodic OBOGS ACBS performance checks.

The performance check shall be based on:

(1) Ideally, a built-in design feature which periodically and automatically assesses oxygen performance and trending in-flight during defined operating conditions; or

(2) An aircraft ground test performed periodically to assess the performance and performance trends of the OBOG system.

If the aircraft engine(s), APU, or ground air source cannot be run, then the OBOG shall be removed and performance checked using a bench test.

The Aircraft Program Office should consult with service OBOGS SMEs to develop and verify maintenance procedures.

A surveillance program should be implemented to assess the effectiveness of:

(1) The interval between the periodic performance checks.

(2) Pass/fail criteria established for OBOGS maintenance practices.

E.3.1.2 Periodic ACBS performance check interval.

Engineering analysis, or maintenance data, or physiologic parameter excursions/events, or any combination thereof, shall be used to determine the interval for conducting the periodic ACBS

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

performance checks. If an analysis or data are unavailable the ACBS shall be tested every 100 flight hours. The intent is to identify low performing ACBS that will impair human performance and physiology. The ACBS performance checks shall include the performance of the breathing regulator, the accuracy of the oxygen monitor, and the accuracy of the pressure sensors.

E.3.1.3 Periodic filter checks.

Engineering analysis, or maintenance data, or physiologic parameter excursions/events, or any combination thereof, shall be used to determine the interval for conducting the periodic filter checks. If an analysis or data are unavailable, the filter checks shall be conducted every 100 flight hours in accordance with the sections below.

E.3.1.4 OBOGS overhaul.

The OBOGS shall be overhauled based on an interval determined by Engineering analysis, or maintenance data, or physiologic parameter excursions/events, or any combination thereof. If an analysis or data are unavailable the overhaul interval shall be every 500-600 flight hours. Molecular sieve canisters shall be replaced during overhaul. The used canisters shall be tested using a method defined and approved by the Government in accordance with E.3.1.6 below or the molecular sieve oxygen generator can be performance checked using a bench test. If a bench check is used the oxygen generator shall be operated at a specified inlet pressure and demand flow. The outlet oxygen concentration can then be compared to standardized performance curves to assess relative performance. The results shall be recorded. Specific tests and pass/fail criteria will vary by aircraft and shall be determined by the program office.

Other critical components (as determined by the Aircraft Program Office) shall also be also be overhauled prior to failure based on a maintenance schedule, which is derived from component reliability data. Critical components shall not be flown until failure.

E.3.1.5 Incident where a chemical contaminant is suspected.

After an aircraft incident where breathing gas contamination is suspected, gas samples should be collected. Gas samples of the breathing gas shall be collected and analyzed for compliance with Table IV. During the sampling, the aircraft engine shall be running. If the samples show the breathing gas is out-of-compliance based in Table IV, the ACBS shall enter the maintenance process. If the ACBS includes OBOGS, the canisters shall be tested based on a method defined in E.3.1.6 below or the oxygen generator can be performance checked using a bench test. The results shall be recorded.

E.3.1.6 Molecular sieve canister assessment methods.

ACBS molecular sieve canisters shall be assessed 1) prior to oxygen generator production; 2) during the ACBS maintenance process; and 3) prior to ACBS overhaul. Manufacturers shall identify one of the methods per E.3.1.6 or a method approved by the program office for testing their molecular sieve canisters. The intent is to identify and analyze molecular sieve canister degradation rates and use this information for trend analysis in order to ensure human physiologic requirements are met. Molecular sieve activity level is a critical parameter for ACBS performance and contaminant protection.

E.3.1.6.1 Molecular sieve canister washout time assessment method.

A method of testing for contaminant permeation is based on measurement of molecular sieve canister washout time. The test is performed by flowing 100% dry oxygen (30 liter/min at a pressure of 75 lb/in^2 gauge) until the outlet gas shows 100% oxygen for a length of time sufficient to confirm the canister is saturated with oxygen - nominally ten (10) minutes. At time zero, the inlet gas is switched from 100% oxygen to 100% nitrogen at the same flow, and the time required for the product gas concentration to drop from 100% to 37% oxygen is recorded. The time on the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

canister under test is compared to the baseline time on an identical canister containing fresh molecular sieve packed in accordance with the manufacturer's specifications and tested at the same temperature and flow rate. The molecular sieve in the test canister can be considered an adequate barrier to contaminant permeation if the time required to reach 37% oxygen concentration remains greater than one-half the baseline time. When the time required for the product gas concentration to drop from 100% to 37% oxygen is less than one-half the baseline, then the canister must be considered marginally effective for contaminant removal and the molecular sieve should be replaced.

E.3.1.6.2 Molecular sieve canister activity assessment method.

This method for assessing the activity of molecular sieve is based on nitrogen adsorption capacity. The amount of nitrogen adsorbed for the test canister is compared to the amount of nitrogen adsorbed on an identical canister filled with fully activated molecular sieve. Any loss in adsorption capacity is due to molecular sieve degradation or loss of available adsorption sites. The method is summarized below. More details may be found in U.S. Patent No. US4916630. The equation used is as follows:

$$Molecular\ Sieve\ Activity\ \% = \frac{(Weight\ of\ Nitrogen\ in\ Test\ Canister(grams) x\ 100)}{Weight\ of\ Nitrogen\ in\ Fully\ Activated\ Canister\ (grams)}$$

E.3.1.6.2.1 Molecular sieve canister activity assessment method additional details.

The void volume in the canister is determined by evacuating the canister and injecting a known amount of helium. The void volume is determined by the change in helium pressure. The canister is evacuated and the initial weight is measured. Pure nitrogen is injected into the canister at a pressure of 60 psig. The canister final weight is measured. Measured Values of pressure, temperature, and weight are used to calculate the weight gain due to nitrogen adsorption. The minimum acceptable activity percentage will be unique for each aircraft and shall be determined by the Aircraft Program Office: 1) minimum acceptable activity percentage for a canister loaded with fresh molecular sieve and 2) minimum acceptable activity percentage for a used but serviceable canister. The manufacturer minimum activity percentages shall be used to determine if the canister passes or fails the molecular sieve activity test.

E.3.1.6.3 Installed performance testing method.

The canisters may be installed into an operating oxygen concentrator (or surrogate) and tested at various pressures and flows to verify the canisters meet acceptable performance criteria. The acceptance criteria shall be agreed to by the program office and contractor.

E.3.1.7 Periodic performance checks.

E.3.1.7.1 Periodic LOX or GOX ACBS performance checks.

Checks shall be periodically conducted to assure the performance of the ACBS in accordance with applicable U.S. Air Force, U.S. Navy or U.S. Marine Corps Technical Orders and Instructions related to the LOX and GOX system and specific aircraft model.

E.3.1.7.2 Periodic ACBS performance checks for ACBS other than OBOGS, LOX or GOX.

Checks of these systems shall be periodically conducted to assure adequate performance. Engineering analysis and/or maintenance data and/or physiologic parameter excursions/events shall be used to determine the interval for conducting the periodic ACBS checks, including breathing regulator, oxygen monitor, and additional ACBS component checks as directed by the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

Aircraft Program Office and coordinated with the relevant subject matter experts. If an analysis or data are unavailable, the checks should be conducted every 100 flight hours.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

APPENDIX F

CALCULATING ALVEOLAR PARTIAL PRESSURE AND OXYGEN CONCENTRATION AT A DESIRED ALTITUDE

F.1 SCOPE

F.1.1 Scope.

This Appendix shows the procedures for calculating the alveolar O_2 partial pressure (P_AO_2) when a person inhales a given O_2 concentration at a particular altitude. It also shows how to determine the O_2 concentration needed for a desired P_AO_2 at a given altitude. The calculations have been used to generate Table F-I. In this table, the Caution column represents normoxia (P_AO_2= 100 mmHg) and the Warning column represents slight hypoxia (P_AO_2 = 85 mmHg). This Appendix also shows how to modify the oxygen schedule to provide the desired minimum P_AO_2 of 30 mmHg in case of rapid decompression. This Appendix is not a mandatory part of the standard. The information contained herein is intended for guidance only.

F.2 APPLICABLE DOCUMENTS

F.2.1 Other Government documents, drawings, and publications.

The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ADVISORY GROUP FOR AEROSPACE RESEARCH & DEVELOPMENT

AGARD-AG-286 Advanced Oxygen Systems for Aircraft, NATO Agardograph 286, John Ernsting and Richard L. Miller (eds.) 1996

(Copies of this document are available online at https://apps.dtic.mil/sti/pdfs/ADA306996.pdf.)

F.3 PROCEDURES FOR CALCULATING THE ALVEOLAR PARTIAL PRESSURE OF OXYGEN

F.3.1 Equations.

The following equations can be used to calculate the P_AO_2:

The alveolar gas equation (found in most textbooks on respiratory physiology) yields P_AO_2:

$$P_AO_2 = \left(P_B - P_{H_2O}\right) \cdot F_IO_2 - P_ACO_2 \left(F_IO_2 + \frac{1-F_IO_2}{R}\right) \qquad \text{(Eq F-1)}$$

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

where:

P_B = cabin barometric pressure plus breathing regulator safety pressure plus PBA (P_B, normally barometric pressure, should be read here as “breathing gas pressure”).

F_IO_2= inspired oxygen fraction = oxygen concentration in volume% / 100

P_ACO_2= alveolar carbon dioxide partial pressure

R = respiratory exchange ratio ($\dot{V}_{CO_2}/\dot{V}_{O_2}$), rate of metabolic CO_2 production divided by rate of metabolic oxygen consumption

P_{H_2O} = water vapor pressure at body temperature

If F_IO_2 = 1 (100% oxygen is inhaled) or if R=1, Eq F-1 simplifies to

$$P_AO_2 = \left(P_B - P_{H_2O}\right) \cdot F_IO_2 - P_ACO_2$$

Equation F-2 can be used to determine what inspired F_IO_2 is needed to generate a desired P_AO_2 at a given altitude:

$$F_IO_2 = \frac{P_AO_2 \cdot R + P_ACO_2}{R \cdot \left(P_B - P_{H_2O} - P_ACO_2\right) + P_ACO_2} \qquad \text{(Eq F-2)}$$

If R = 1, Eq F-2 simplifies to $F_IO_2 = (P_AO_2 + P_ACO_2) / (P_B - P_{H_2O})$.

F.3.2 Step-by-step procedures.

The following are the step-by-step procedures to calculate alveolar partial pressure of oxygen.

F.3.2.1 Calculating P_AO_2 at different altitudes.

Step 1: Choose values for:

P_ACO_2: for normocapnia 40 mmHg, 53 hPa.

R: varies with diet, range of 0.7 to 1.0. 0.83, 0.85 or 1 are commonly assumed.

P_{H_2O}: 47 mmHg, 63 hPa at body temperature, independent of altitude.

F_IO_2: 0.2095 if air is breathed, 1 if 100% O_2 is breathed.

NOTE: R and P_ACO_2 are approximations and will depend on a person’s normal hypoxic response (which is hyperventilation of varying degrees) and any changes in the person’s breathing induced by the ACBS.

Step 2: Determine the ambient pressure for a desired cabin altitude.

Step 3: Use Equation F-1 to calculate P_AO_2.

Step 4: Repeat steps 2 and 3 for the desired range of altitudes.

Step 5: Tabulate and/or graph the results as desired.

F.3.2.2 Calculating F_IO_2 at different altitudes.

Step 1: Set the desired values for P_ACO_2.

Step 2: Determine the ambient pressure for a cabin altitude.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

Step 3: Use Equation F-2 to calculate F_IO_2.

Step 4: Repeat steps 2 and 3 for the desired range of altitudes.

Step 5: Tabulate and/or graph the results as desired.

NOTE on Table F-I below: The column previously labelled in MIL-STD-3050 as “caution” has been deleted to match the labels on Figures 2a and 2b. The column previously labelled as “normoxia” is now labelled as Caution.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

TABLE F- I. Oxygen vs altitude schedule using alveolar gas equation[1/].

Cabin altitude		Pressures				P_IO_2 from cabin[1/] (mmHg)	Design Zone (% O_2)		Caution[2/]	Caution [3/]	Warning
(ft)	(m)[4/]	(atm abs)	(mmHg)	(inHg)	(hPa)		max	min	(% O_2)	(% O_2)	(% O_2)
0	0	1.00	760	29.92	1013	149	60	40	20.5	20.5	18.5
500	152	0.98	746.4	29.38	994.8	147	60	40	20.9	20.9	18.8
1,000	305	0.96	732.9	28.85	976.9	144	60	40	21.3	21.3	19.2
1,500	457	0.95	719.7	28.33	959.3	141	60	40	21.8	21.8	19.6
2,000	610	0.93	706.7	27.82	941.9	138	60	40	22.2	22.2	19.9
2,500	762	0.91	693.8	27.31	924.8	136	60	40	22.6	22.6	20.3
3,000	914	0.90	681.2	26.82	907.9	133	60	40	23.1	23.1	20.7
3,500	1067	0.88	668.7	26.32	891.3	130	60	40	23.5	23.5	21.1
4,000	1219	0.86	656.4	25.84	874.9	128	60	40	24.0	24.0	21.6
4,500	1372	0.85	644.3	25.36	858.8	125	60	40	24.5	24.5	22.0
5,000	1524	0.83	632.4	24.89	842.9	123	60	40	25.0	25.0	22.4
5,500	1676	0.82	620.7	24.43	827.2	120	60	40	25.5	25.5	22.9
6,000	1829	0.80	609.1	23.98	811.8	118	60	40	26.0	26.0	23.4
6,500	1981	0.79	597.7	23.53	796.6	115	60	40	26.5	26.5	23.8
7,000	2134	0.77	586.5	23.09	781.7	113	60	40	27.1	27.1	24.3
7,500	2286	0.76	575.5	22.65	766.9	111	60	40	27.6	27.6	24.8
8,000	2438	0.74	564.6	22.22	752.4	108	60	40	28.2	28.2	25.3
8,500	2591	0.73	553.9	21.80	738.2	106	60	40	28.8	28.8	25.9
9,000	2743	0.71	543.3	21.39	724.1	104	60	40	29.4	29.4	26.4
9,500	2896	0.70	533.0	20.98	710.3	102	60	40	30.0	30.0	27.0
10,000	3048	0.69	522.8	20.58	696.6	100	60	40	30.6	30.6	27.5
10,500	3200	0.67	512.7	20.18	683.2	98	60	40	31.3	31.3	28.1
11,000	3353	0.66	502.8	19.79	670.0	95	60	40	31.9	31.9	28.7
11,500	3505	0.65	493.1	19.41	657.0	93	60	40.4	32.6	32.6	29.3
12,000	3658	0.64	483.5	19.03	644.2	91	60	41.2	33.3	33.3	30.0
12,500	3810	0.62	474.0	18.66	631.7	89	60	42.2	34.0	34.0	30.6
13,000	3962	0.61	464.8	18.29	619.3	88	60	43.1	34.8	34.8	31.3
13,500	4115	0.60	455.6	17.93	607.1	86	60	44.1	35.6	35.6	32.0
14,000	4267	0.59	446.6	17.58	595.1	84	60	45.0	36.3	36.3	32.7
14,500	4420	0.58	437.8	17.23	583.3	82	60	46.1	37.1	37.1	33.4
15,000	4572	0.56	429.1	16.88	571.7	80	60	47.1	38.0	38.0	34.1
15,500	4724	0.55	420.5	16.55	560.2	78	61.5	48.2	38.8	38.8	34.9
16,000	4877	0.54	412.1	16.22	549.0	76	63.0	49.3	39.7	39.7	35.7
16,500	5029	0.53	403.8	15.89	538.0	75	64.5	50.4	40.6	40.1	36.5
17,000	5182	0.52	395.7	15.57	527.1	73	66.0	51.6	41.5	41.7	37.3
17,500	5334	0.51	387.7	15.25	516.4	71	67.5	52.8	42.5	43.8	38.2
18,000	5486	0.50	379.8	14.94	505.9	70	69.0	54.1	43.5	45.8	39.1
18,500	5639	0.49	372.0	14.64	495.5	68	70.5	55.4	44.5	47.8	40.0
19,000	5791	0.48	364.4	14.34	485.4	66	72.0	56.7	45.5	49.9	40.9
19,500	5944	0.47	356.9	14.04	475.3	65	73.5	58.1	46.6	51.9	41.9
20,000	6096	0.46	349.5	13.75	465.5	63	100	59.5	47.7	54.0	42.9
20,500	6248	0.45	342.3	13.46	455.8	62	100	61.0	48.8	56.0	43.9
21,000	6401	0.44	335.2	13.18	446.3	60	100	62.5	50.0	56.0	44.9
21,500	6553	0.43	328.2	12.91	437.0	59	100	64.0	51.2	56.0	46.0

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

TABLE F- I. Oxygen vs altitude schedule using Alveolar Gas Equation[1/] - Continued.

Cabin altitude		Pressures				P_IO_2 from cabin[1/]	Design Zone (% O_2)		Caution [2/]	Caution [3/]	Warning
(ft)	(m)[4/]	(atm abs)	(mmHg)	(inHg)	(hPa)	(mmHg)	max	min	(% O_2)	(% O_2)	(% O_2)
22,000	6706	0.42	321.3	12.64	427.8	57	100	65.6	52.5	56.0	47.2
22,500	6858	0.41	314.5	12.37	418.8	56	100	67.3	53.8	56.0	48.3
23,000	7010	0.40	307.9	12.11	409.9	55	100	69.0	55.1	56.0	49.5
23,500	7163	0.40	301.3	11.85	401.2	53	100	70.8	56.4	56.4	50.7
24,000	7315	0.39	294.9	11.60	392.6	52	100	72.6	57.9	57.9	52.0
24,500	7468	0.38	288.6	11.35	384.2	51	100	74.5	59.3	59.3	53.3
25,000	7620	0.37	282.4	11.10	375.9	49	100	94	60.8	60.8	54.7
25,500	7772	0.36	276.3	10.86	367.8	48	100	94	62.4	62.4	56.1
26,000	7925	0.36	270.3	10.63	359.8	47	100	94	64.0	64.0	57.5
26,500	8077	0.35	264.4	10.40	351.9	46	100	94	65.7	65.7	59.0
27,000	8230	0.34	258.7	10.17	344.2	44	100	94	67.4	67.4	60.6
27,500	8382	0.33	253.0	9.94	336.7	43	100	94	69.2	69.2	62.2
28,000	8534	0.33	247.4	9.72	329.2	42	100	94	71.0	71.0	63.8
28,500	8687	0.32	242.0	9.51	321.9	41	100	94	72.9	72.9	65.6
29,000	8839	0.31	236.6	9.30	314.8	40	100	94	74.9	74.9	67.3
29,500	8992	0.30	231.1	9.09	307.7	39	100	94	77.1	77.1	69.3
30,000	9144	0.30	226.1	8.88	300.8	38	100	94	79.1	79.1	71.1
30,500	9296	0.29	221.1	8.68	294.0	36	100	94	81.3	81.3	73.1
31,000	9449	0.28	216.1	8.49	287.4	35	100	94	83.6	83.6	75.1
31,500	9601	0.28	211.2	8.29	280.8	34	100	94	86.0	86.0	77.3
32,000	9754	0.27	206.4	8.11	274.4	33	100	94	88.5	88.5	79.5
32,500	9906	0.26	201.6	7.92	268.1	32	100	94	91.0	91.0	81.8
33,000	10058	0.26	197.0	7.74	261.9	31	100	94	93.0	93.0	84.2
33,500	10211	0.25	192.5	7.56	255.9	30	100	94	93.0	93.0	86.7
34,000	10363	0.25	188.0	7.38	249.9	30	100	94	93.0	93.0	89.3
34,500	10516	0.24	183.6	7.21	244.1	29	100	94	93.0	93.0	92.0
35,000	10668	0.24	179.3	7.04	238.4	28	100	94	-		93
36,000	10973	0.22	171.0	6.71	227.2	26	100	94	-		93
37,000	11278	0.21	163.0	6.40	216.5	24	100	94	-		93
38,000	11582	0.20	155.4	6.09	206.2	23	100	94	-		93
39,000	11887	0.194	148.1	5.80	196.3	21	100	94	-		93
40,000	12192	0.184	141.2	5.52	186.8	20	100	94	-		93
41,000	12497	0.175	134.6	5.25	177.7	18	100	94	-		93
42,000	12802	0.167	128.3	4.99	169.0	17	100	94	-		93
43,000	13106	0.158	122.3	4.74	160.5	16	100	94	-		93
44,000	13411	0.151	116.6	4.50	152.5	15	100	94	-		93
45,000	13716	0.143	111.1	4.28	144.7	13	100	94	-		93
46,000	14021	0.136	105.9	4.06	137.3	12	100	94	-		93
47,000	14326	0.129	101.0	3.85	130.2	11	100	94	-		93
48,000	14630	0.122	96.27	3.65	123.4	10	100	94	-		93
49,000	14935	0.115	91.77	3.45	116.9	9	100	94	-		93
50,000	15240	0.109	87.49	3.27	110.7	8	100	94	-		93

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

NOTES:

1/ $P_IO_2 = F_IO_2 \cdot (P_B - P_{H2O})$ is the O_2 partial pressure in humidified gas at body temperature at a given altitude.

2/ For aircraft with decompression times greater than 1 second (such as cargo transport aircraft) (see Figure 2a). The calculations for caution (normoxia) are based on the alveolar gas equation with P_ACO_2 = 40 mmHg and R = 0.83. Normoxia is defined as P_AO_2 = 100 mmHg.

3/ For aircraft with decompression time less than or equal to one (1) second (such as aircraft with high probability of rapid decompression, e.g., fighter/attack aircraft) (see Figure 2b). The calculations for caution (normoxia) are based on the alveolar gas equation with P_ACO_2 = 40 mmHg and R = 0.83. Normoxia is defined as P_AO_2 = 100 mmHg.

4/ Altitude in meters is a direct conversion of altitude in feet, which is expressed to the nearest 500 ft (152 m). No higher precision should be attributed to the values in meters.

F.4 PROCEDURE TO MODIFY THE CAUTION LEVEL IN THE OXYGEN SCHEDULE TO PROTECT DURING RAPID DECOMPRESSION

F.4.1 Procedure.

The caution level in the oxygen schedule is generally the oxygen fraction that provides normoxic gas. However, for aircraft that may experience rapid decompression, “Ernsting’s Notch” increases that oxygen fraction at high cabin altitude. The procedure for determining the modification follows Ernsting’s scheme, outlined in AGARD-AG-286: “Assuming that 100% oxygen is delivered to the respiratory tract immediately when the decompression occurs, the alveolar PO_2 can be prevented from falling below 30 mmHg by ensuring that the gas breathed before the decompression contains an adequate concentration of oxygen and that the intrapulmonary pressure does not fall below 115-120 mmHg absolute.”

F.4.2 Step-by-step procedure.

Step-by-step procedure for determining the oxygen concentrations needed are as follows:

Step 1: Determine the atmospheric pressure at the maximum altitude for the aircraft:

Step 2: Add the pressure difference that is normally maintained; this gives the cabin pressure at maximum aircraft altitude;

Step 3: Determine the P_AO_2 during normal operation that is needed to give a P_AO_2 of 30 mmHg after decompression. This would be 30 x pressure ratio. The pressure ratio is calculated as ratio of the pressure of the breathing gas minus the water vapor pressure before decompression divided by the pressure of the breathing gas minus the water vapor pressure after decompression. From this and the pressure of the breathing gas during normal operation, calculate F_IO_2 using Equation F-2.

Step 4: Determine the cabin altitude at which the oxygen fraction for normoxia corresponds to that for P_AO_2 = 30 mmHg after rapid decompression. This depends on the pressurization schedule of the aircraft, but is 16,000 ft. in an aircraft with a 5 psig pressure differential. Taper the oxygen concentration to the sea level equivalence (normoxia) line at this cabin altitude linearly to meet the oxygen concentration and altitude determined in Step 3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

F.4.3 Example.

Assume that an aircraft flying at 50,000 feet has a 5 psig pressure differential during normal operations. Its ACBS adds up to 30 mmHg of PBA (starting at cabin altitudes of 40,000 feet, values from AGARD-AG-286). The value for P_B to use in the alveolar equation is the pressure of the breathing gas, that is, cabin pressure + any positive pressure from the life support system. For instance, if the cabin is at 50,000 ft. then the P_B to use is 87.5 mmHg + 30 mmHg = 117.5 mmHg.

Step 1: In case of decompression at maximum aircraft altitude, the cabin pressure becomes the barometric pressure at 50,000 feet, which is 1.69 psia (87.5 mmHg) (see Table F-I). The positive pressure from the ACBS delivers breathing gas at 87.5 mmHg + 30 mmHg = 117.5 mmHg. Eq. F-2 with 117.5 mmHg substituted for P_B and 30 mmHg for P_AO_2, F_IO_2 = 1 (100% oxygen) shows that the alveolar oxygen pressure becomes the minimum desired 30 mmHg.

Step 2: With normal pressure control, the cabin pressure at maximum aircraft altitude is 1.69 psia + 5 psig = 6.69 psia (346 mmHg). This corresponds to cabin altitude between 20,000 and 20,500 feet (see Table F-I).

Step 3: The pressure ratio is (346 - 47) / (117.5 – 47) = 4.2. Thus, the target (minimum acceptable) P_AO_2 with normal pressure control would be 4.2 x 30 = 126 mmHg. From Eq. F-2, assuming that R = 0.83, the oxygen concentration needs to be 57% to achieve this alveolar partial pressure at a cabin altitude of 20,000 feet. From Ernsting's calculation, the value was 56%. If the value for R was 0.9 or 1 (also realistic), then the value becomes 56%. To maintain the historical precedence, the value of 56% will remain. The notch exception ends at 20,000 feet, at higher altitudes the normal schedule applies (hence the horizontal line).

Step 4: The barometric pressure at 40,000 feet is 2.73 psia, and, in the case of rapid decompression, this is the lowest altitude at which PBA would be supplied for the aircraft in this example. With normal pressure control the cabin pressure is 7.73 psia, corresponding to 16,500 feet. For a 5 psig pressure differential, Ernsting calculated a modified oxygen schedule (see AGARD-AG-286), increasing the oxygen concentration above that of the regular schedule for cabin altitudes between 16,000 and 20,000 feet (see Table F-I). This is the zone with PBA in event of rapid decompression.

Table F-II shows a modification of the caution level of the oxygen schedule in Table F-I for an aircraft like that in the example, one capable of flying at 50,000 feet (aircraft altitude) and with a 5 psig pressure differential and an ACBS that provides PBA = 30 mmHg at a cabin altitude of 50,000 feet. Figure F-1 illustrates the values shown in Table F-II. The dotted line between 16,000 feet and 20,000 feet represents "sea level equivalent", that is, normoxia. The, solid line in that area shows the modified schedule.

For this example aircraft, cabin altitudes between about 20,000 and 50,000 feet represent a loss of pressure control within the altitudes where the aircraft can fly. For cabin altitudes above about 20,000 feet, the breathing gas must be 100% oxygen delivered with a positive pressure (PBA) of up to 30 mmHg.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

TABLE F- II. Modified Table F- I oxygen schedule needed to prevent hypoxia following decompression.

Cabin Altitude (ft.)	Cabin Altitude (m)1/	O_2 (%)
16,000	4,877	39.7
16,500	5,029	41.7
17,000	5,182	43.8
17,500	5,334	45.8
18,000	5,486	47.9
18,500	5,639	49.9
19,000	5,791	52.0
19,500	5,944	54.0
20,000	6,096	56.1

NOTES: 1/ Altitude in meters is a direct conversion of altitude in feet, which is expressed to the nearest 500 ft. (152 m). No higher precision should be attributed to the values in meters.

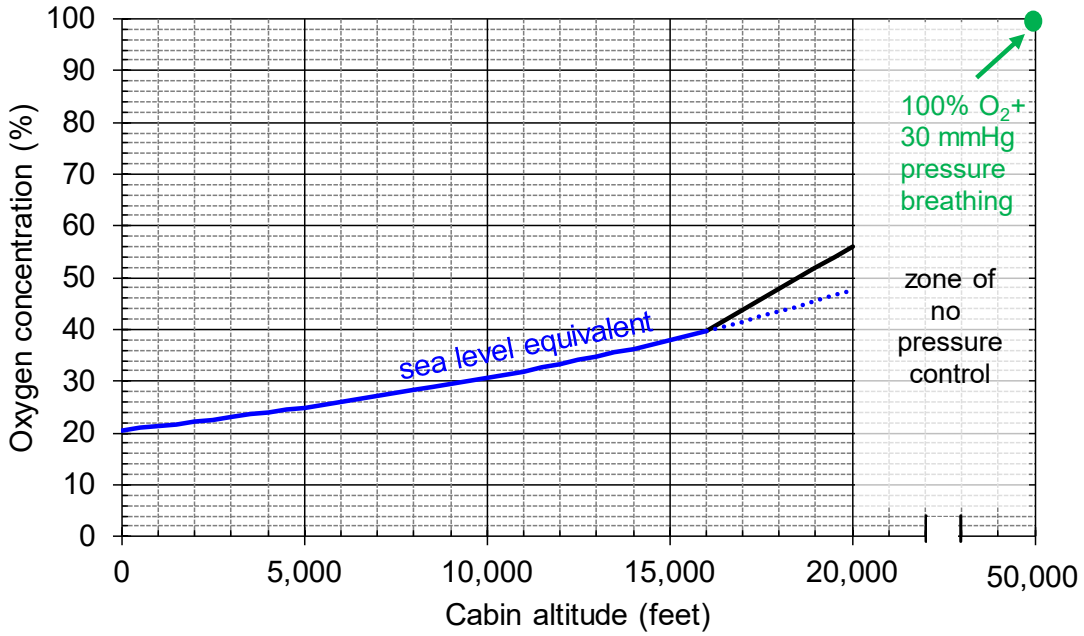


FIGURE F-1. Illustration of modified oxygen schedule needed to prevent hypoxia following decompression for aircraft with a cabin differential of 5 psi using a regulator with max pressure of 30 mmHg at 50,000 ft. with pressure onset at 39,000 ft.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

APPENDIX G

SUPPLEMENTAL INFORMATION USED TO ESTABLISH ACBS REQUIREMENTS

G.1 SCOPE

G.1.1 Scope.

This Appendix contains additional guidance documents, which may be used for guidance during the design, acquisition, certification or sustainment/maintenance for an ACBS. This Appendix is not a mandatory part of the standard. The information contained herein is intended for guidance only.

G.2 DOCUMENTS USED FOR GUIDANCE

G.2.1 Guidance documents.

The following documents may be used for guidance.

FIVE EYES AIR FORCE INTEROPERABILITY COUNCIL

ADV PUB ASMG 4060	The Minimum Quality Criteria for On-Board Generated Oxygen
AIR STD AMD 4068	Physiological Requirements for Aircrew Oxygen Masks for Use at High Breathing Pressures (implemented by 3050)
AIR STD ASM 4038	Physiological Evaluation of Aircraft Oxygen Delivery Systems
AIR STD ASM 4039	Minimum Physiological Requirements for Aircrew Demand Breathing Systems

DEPARTMENT OF DEFENSE SPECIFICATIONS

JSSG 2010-10	Crew Systems Oxygen Systems Handbook
MIL-D-19326	Design and Installation of Liquid Oxygen Systems in Aircraft, General Specification For

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-464	Electromagnetic Environmental Effects Requirements for Systems
MIL-STD-882	System Safety
MIL-STD-1472	Human Engineering
MIL-STD-1760	Aircraft/Store Electrical Interconnection System

(Copies of these documents are available online at https://quicksearch.dla.mil.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

AIR FORCE MANUAL

AFMAN 11-202V3	Flight Operations

AIR FORCE PAMPHLET

AFPAM 161-18	Flight Surgeon's Guide (Table 2-2) 1978

(Copies of these documents are available online at www.e-publishing.af.mil/.)

AIR FORCE TECHNICAL ORDER (TO)

TO 00-20-1	Aerospace Equipment Maintenance Inspection, Documentation, Policies, and Procedures
TO 15X-1-1	Maintenance Instruction – Oxygen Equipment
TO 42B6-1-1	Technical Manual – Quality Control Aviators Breathing Oxygen

(Copies of these documents are available by accessing the Enhanced Technical Information Management System (ETIMS) through the AF Portal at https://www.my.af.mil/. Open the Applications A-Z Index and go to ETIMS.)

DEFENSE TECHNICAL INFORMATION CENTER (DTIC)

AFRL-SA-WR-SR-2011-0003	Handbook of Aerospace and Operational Physiology; Andrew D. Woodrow, James T. Webb (ADA548404)
AFRL-SA-WP-SR-2016-0018	Handbook of Aerospace and Operational Physiology, 2nd Edition, Appendix 1A; R. W. Maresh; Andrew D. Woodrow; James T. Webb (AD1020889)
	Development of Comprehensive Performance Standards for Underwater Breathing Apparatus; Warkander, Dan E., Lundgren, Claes E.G. (ADA487859)
	U.S. Standard Atmosphere, 1962 (AD0659893)

(Copies of these documents are available online at https://search.dtic.mil/.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodians:
Navy - AS
Air Force - 11

Preparing activity:
Air Force - 11

Review activities:
Air Force - 01, 06, 10, 68

(Project 15GP-2019-016)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:01Z
Check the source to verify that this is the current version before use.